

Master Diploma Examination¹

Revision Notes—Physics (Studies in English)

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???

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¹Translated into English with some modifications from an analogous text of *Krokosz et al.*

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Check that all section content actually answers the questions.

Hydrogen atom, fine hyperfine should be somewhere.

somewhere write the band of optical light in hz nad nm

1 Inertial frames, principles of relativity, the Galilean transformation, the Lorentz transformation and its consequences

An inertial frame is a frame of reference with respect to which every body, not subject to external interactions with other bodies, moves in uniform rectilinear motion or remains at rest. The existence of such a frame is postulated by Newton's first law of motion.

The Galilean principle of relativity states that the laws of motion are identical in all inertial frames of reference, i.e. there is no privileged frame of reference. It was formulated in 1604 in the form of the following law: all frames of reference moving with respect to one another at constant velocity (including $\vec{v} = 0$) are equivalent. Its consequence is the set of Galilean transformations between two frames of reference (t, x, y, z) and (t', x', y', z') with the primed one moving with velocity v with respect to the unprimed one along the x axis:

$$\begin{aligned}t' &= t \\x' &= x - vt \\y' &= y \\z' &= z\end{aligned}\tag{1}$$

Einstein additionally postulated that in every inertial frame of reference the speed of light has the same value. He made this assumption in 1905 based on an analysis of the transformations of Maxwell's equations, where he found that the electric and magnetic fields are not separately fixed: what one observer describes as a purely electric field, another observer moving with respect to the first describes as a mixture of an electric and a magnetic field. The two are aspects of one object rather than two independent fields.

The Galilean transformations were thereby generalised to the Lorentz transformation to accommodate the Einstein's postulate:

$$\boxed{\begin{aligned}t' &= \gamma (t - xv/c^2) \\x' &= \gamma (x - tv) \\y' &= y \\z' &= z\end{aligned}}\tag{2}$$

where the parameter γ is known as the Lorentz factor:

$$\gamma = \frac{1}{\sqrt{1 - \beta^2}} \geq 1, \quad \beta \equiv \frac{v}{c}.\tag{3}$$

Note that in the limit $v/c \rightarrow 0$ we have $\gamma \rightarrow 1$ and the Lorentz transformation reduces to the Galilean one (Eq. 1).

The consequences of the Lorentz transformation include:

1. Time dilation, described by the formula $\Delta t = \gamma \Delta t_0$, where Δt_0 is the *proper time*—time recorded in the rest frame of the timed event, and Δt is the time recorded by another observer moving with velocity v with respect to the first. An interval is *shortest* in the frame in which it is at rest.
2. Lorentz contraction, described by the formula $L = L_0/\gamma$, where L_0 is the *proper length*—length of an object at rest, and L is the length of the object recorded by an observer moving with respect to it at velocity v . An object is *longest* in the frame in which it is at rest.
3. Relativity of simultaneity: two events determined by one observer to be simultaneous need not be simultaneous for another observer. Simultaneity translates to other frames only for the events that happen at the same point in space.
4. The description of physical quantities in four dimensions. Since [Eq. 2](#) mixes t with x , a split into a ‘time part’ and a ‘space part’ is not something all observers agree on. Related quantities are therefore collected into *four-vectors*, whose four components transform into one another exactly as position does,

$$x^\mu = \begin{bmatrix} ct \\ x \\ y \\ z \end{bmatrix}, \quad p^\mu = \begin{bmatrix} E/c \\ p_x \\ p_y \\ p_z \end{bmatrix}.$$

Position is the prototype, and there the zeroth slot is literally time. What makes any other set of four quantities a four-vector is not an analogy with position but a transformation law. An equation written as an equality of four-vectors automatically holds in every inertial frame, because both sides transform in the same way. And each four-vector has one combination of its components that every observer agrees on: for position it is the spacetime interval $(c^2t^2 - X^2)$, and for momentum it is $(E^2 - p^2c^2)$.

Mass–energy.

Writing out the invariant of the four-momentum gives the *energy–momentum relation*

$$\boxed{E^2 = p^2c^2 + m^2c^4}, \quad (4)$$

where E is the total energy of a free body, p the magnitude of its momentum, and m its *invariant mass*, meaning the mass measured in the frame in which the body is at rest. Two limits carry most of the physics. For a body at rest we have $p = 0$ and the relation collapses to Einstein’s $E = mc^2$, so that mass is itself a store of energy at the conversion rate $c^2 \simeq 9 \times 10^{16} \text{ J/kg}$ —the reason a milligram of mass defect suffices to power a nuclear reaction. For a massless particle such as the photon we have $m = 0$ and the relation gives $E = pc$, a case $E = mc^2$ cannot describe at all. The invariant mass of a composite system also counts the internal energy of its parts, so a hot or compressed body is very slightly more massive than a cold relaxed one, and roughly 99% of a proton’s mass is the kinetic energy of its quarks and the energy of the gluon field rather than the quark masses themselves.

2 Fundamental interactions; carriers and range of interactions, charges and characteristic energy scales

Every process in physics comes down to particles interacting, and every known interaction belongs to one of four families. In a quantum theory an interaction is not a force reaching across empty space but an *exchange*: two particles interact by passing a third particle between them, called the *interaction carrier*. The properties of the carrier—its mass, its spin, and the kind of charge it responds to—fix everything else, in particular how far the interaction reaches and how strong it is.

What each interaction is responsible for:

- **Strong.** Binds quarks into protons, neutrons and other *hadrons*, meaning particles built out of quarks. What is left over from this binding reaches slightly beyond each nucleon and holds protons and neutrons together in the nucleus. That residue is what overcomes the electrostatic repulsion between the protons and lets nuclei with more than one proton exist at all.
- **Electromagnetic.** Binds electrons to nuclei, atoms into molecules and molecules into solids, so all of chemistry and all of ordinary matter's rigidity. Light is a propagating disturbance of the same field.
- **Weak.** Converts one kind of quark or lepton into another. It drives beta decay, the first step of hydrogen fusion in stars, and essentially every interaction a neutrino ever has.
- **Gravitational.** Acts on everything that carries energy. It is by far the feeblest between two individual particles, but it dominates on astronomical scales because its charge—mass-energy—comes with one sign only and never cancels.

The first three are described by the *Standard Model*, the quantum field theory of elementary particles. Matter in it consists of twelve spin-1/2 particles arranged in three generations: six quarks (up and down, charm and strange, top and bottom) and six leptons (electron, muon and tau, each with its neutrino). Ordinary matter uses only the first generation—up and down quarks and the electron—while the heavier generations are copies that decay quickly. The interactions are carried by spin-1 bosons: eight gluons, the photon, and the W^+ , W^- and Z^0 bosons. One further particle, the spin-0 *Higgs boson* of mass about 125 GeV, is the quantum of the field that gives the W , the Z and the charged fermions their masses. Gravitation stands outside this framework and has no accepted quantum description.

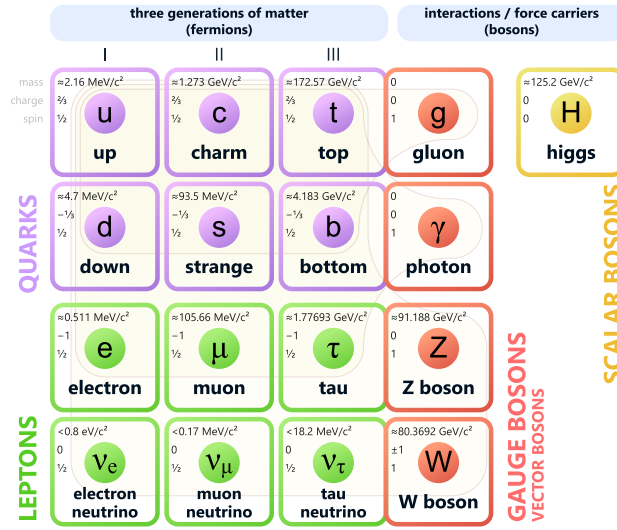


Figure 1: Standard Model of Elementary Particles.

interaction	carrier	charge	range	strength
strong	8 gluons, massless	colour	$\sim 10^{-15}$ m	1
electromagnetic	photon, massless	electric	∞	$\sim 10^{-2}$
weak	$W^\pm$ (80 GeV), Z^0 (91 GeV)	weak isospin	$\sim 10^{-18}$ m	$\sim 10^{-6}$
gravitational	graviton, hypothetical	mass–energy	∞	$\sim 10^{-39}$

Table 1: The four interactions. All carriers have spin 1 except the graviton, which would have spin 2. Strengths are orders of magnitude only, for two protons about 10^{-15} m apart, and different textbooks quote somewhat different figures because the comparison itself is not unique.

A particle feels an interaction only if it carries the matching charge, which is what the third column lists. Colour charge comes in three values conventionally called red, green and blue, and is carried by quarks and gluons alone. Weak isospin is the charge of the weak interaction and is carried by all quarks and leptons.¹ Its name records a mathematical analogy—it obeys the same algebra as spin—but it is not intrinsic angular momentum and has nothing to do with rotation in space. It sits in the charge column because that is the role it plays: it is the label deciding whether a particle couples to the interaction at all. A neutrino has neither electric charge nor colour, which is why it interacts only weakly and passes through the Earth almost undisturbed. Carriers may themselves be charged: each gluon carries one colour and one anticolour, so gluons attract one another, whereas the photon is electrically neutral and ignores other photons. This self-attraction of the gluon field is the reason the strong interaction behaves so differently from the other three.

¹More precisely by their left-handed components, which is why the W couples only to those.

Energies are quoted in *electronvolts*, 1 eV being the energy an electron gains across a potential difference of one volt, $1 \text{ eV} = 1.602 \times 10^{-19} \text{ J}$, with the multiples keV, MeV and GeV. Masses are quoted in the same units by *mass–energy equivalence*, $E = mc^2$, a mass being named by the rest energy it corresponds to. Strictly the unit is then eV/c^2 , but the c^2 is left out and one simply says the W boson has a mass of 80 GeV. In SI units $1 \text{ eV}/c^2 = 1.783 \times 10^{-36} \text{ kg}$.

The ranges in [Table 1](#) are not independent facts but follow from the carrier masses. The energy–time uncertainty relation $\Delta E \Delta t \geq \hbar/2$ says that a state existing only briefly cannot have a sharply defined energy. Note what this is about: the energy spread of a short-lived state itself, not the precision of our instruments and not the mass of the particle species, which is a fixed number measured on real particles in accelerators. An exchanged carrier is never observed directly and so need not balance the energy books—it is *virtual*, in effect borrowed from the vacuum provided the loan is repaid quickly. Borrowing at least the rest energy mc^2 limits its life to $\Delta t \sim \hbar/mc^2$, and travelling at most at the speed of light it covers

$$\boxed{R \sim c \Delta t \sim \frac{\hbar}{mc}}, \quad (5)$$

the Compton wavelength of the carrier. A heavy carrier therefore gives a short-ranged interaction and a massless one an unlimited range, which is why electromagnetism and gravitation reach as far as they do. Using $\hbar c \approx 197 \text{ MeV fm}$ with a carrier rest energy of 80 GeV gives $R \approx 2.5 \times 10^{-18} \text{ m}$ for the weak interaction, the order of magnitude quoted in [Table 1](#).

The strong interaction is the apparent exception, since its carrier is massless and yet it does not reach beyond the nucleus. Its range is set instead by *confinement*: because gluons attract one another, colour charge can never be isolated, and gluons do not travel freely between well-separated particles. What acts between nucleons is a residual effect carried by *pions*, the lightest hadrons, of rest energy about 140 MeV, and [Eq. 5](#) then returns the familiar $1.4 \times 10^{-15} \text{ m}$.

Beta decay shows the pieces working together. A down quark inside a neutron turns into an up quark and emits a W^- , which decays at once into an electron and an electron antineutrino, so that overall $n \rightarrow p + e^- + \bar{\nu}_e$. The energy released is a few MeV, thousands of times less than the 80 GeV needed to make a real W . The boson can therefore only ever be virtual, and it is this mismatch of scales—not any small coupling—that makes the weak interaction weak and slow at everyday energies.

Collecting these, each interaction has its own characteristic energy scale:

- **Electromagnetic.** Atomic and molecular binding, from about 1 eV for a chemical bond to 13.6 eV to ionise hydrogen, rising to keV for inner electrons of heavy atoms.
- **Strong.** Nuclear binding of roughly 8 MeV per nucleon, hadron masses of order 1 GeV, and an underlying scale of a few hundred MeV at which the interaction becomes strong.

- **Weak.** Set by the carrier masses, of order 100 GeV, even though the energy actually released in weak decays is only of order MeV.
- **Gravitational.** No scale at all in everyday physics, since the interaction has no intrinsic energy of its own. Quantum effects are expected only at the *Planck energy*, about 10^{19} GeV, which is beyond any conceivable accelerator.

3 Conservation laws in physics and their connection with symmetry laws

The *symmetries* of a physical system are closely connected with its *conservation laws*. A symmetry here means that the system goes on behaving in exactly the same way after some transformation has been applied to it—shifting the whole experiment in time, moving it to another place, turning it to face another direction.

Noether’s theorem formalises the connection: to every continuous symmetry of the action there corresponds a conserved quantity. *Continuous* means the transformation can be made arbitrarily small, as a rotation can but a mirror reflection cannot.

3.1 The Lagrangian, the action and the equation of motion

The *Lagrangian* of a system is the difference between its kinetic and its potential energy,

$$\mathcal{L} \equiv T - V, \tag{6}$$

and the *action* is the Lagrangian accumulated along a path in time,

$$S \equiv \int_{t_1}^{t_2} \mathcal{L} dt. \tag{7}$$

S is not a number belonging to the system but a number belonging to a whole candidate path, one value for every path one might consider. An object that eats a function and returns a number in this way is called a *functional* (Section 7).

Demanding that S be *stationary*—unchanged to first order when the path is deformed slightly with its two endpoints held fixed—singles out the path actually followed and yields the *Euler–Lagrange equation*,

$$\boxed{\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}} \right) = \frac{\partial \mathcal{L}}{\partial q}}, \tag{8}$$

where q is any coordinate used to specify the configuration of the system and $\dot{q} = dq/dt$. This is the principle of least action (Section 7) in one line, and for a single particle in a potential it returns Newton's second law (Eq. 11).

The two derivatives² in Eq. 8 do different jobs. The Lagrangian is treated as a function of three independent slots, $\mathcal{L}(q, \dot{q}, t)$, so the partial derivative $\partial/\partial q$ varies q alone and holds $\dot{q}$ and t fixed. The total derivative d/dt is taken after an actual trajectory has been substituted, where q and $\dot{q}$ are themselves functions of time, so by the chain rule it collects every dependence on t , the implicit ones as well as the explicit one.

3.2 Noether's theorem

If the action of a system is invariant under a continuous transformation, then there is a quantity built from the state of the system that does not change in time.

The theorem is stated for the action rather than for the equations of motion because the action is what those equations follow from, so a symmetry of S is automatically a symmetry of the dynamics. In practice the symmetry is tested on $\mathcal{L}$ instead, which is easier: S is only $\mathcal{L}$ integrated over time, so a transformation leaving $\mathcal{L}$ unchanged leaves S unchanged too.³ Each example below therefore begins by asking which variable is missing from the Lagrangian.

3.3 Examples of the application of Noether's theorem

3.3.1 Conservation of energy

Symmetry under a shift in time (homogeneity of time) leads to the law of conservation of energy. Suppose the Lagrangian does not depend explicitly on time, meaning that repeating the experiment tomorrow changes nothing,

$$\frac{\partial \mathcal{L}}{\partial t} = 0.$$

Differentiating $\mathcal{L}$ along the trajectory and substituting Eq. 8 then gives $d/dt (\dot{q} \partial \mathcal{L} / \partial \dot{q} - \mathcal{L}) = -\partial \mathcal{L} / \partial t = 0$, so that

²A *partial derivative* $\partial f / \partial x$ is the rate of change of f with respect to x when every other argument of f is held fixed, so its value depends entirely on which variables those are — thermodynamics makes this explicit by writing the held quantities as subscripts, as in $(\partial U / \partial V)_T$. A *total derivative* df/dt asks a different question: how f changes along an actual path, counting every route by which t influences it. The *chain rule* connects the two,

$$\boxed{\frac{df}{dt} = \frac{\partial f}{\partial t} + \sum_i \frac{\partial f}{\partial q_i} \frac{dq_i}{dt}}, \quad (9)$$

for a function $f(q_1, \dots, q_n, t)$ whose arguments q_i themselves vary with t . The first term is the *explicit* dependence on t — what changes because the rule defining f was written with a t in it — and the sum is the *implicit* dependence, what changes because the system has moved. This distinction is the whole content of several standard results.

³The converse fails. $\mathcal{L}$ may change by the total time derivative of some function and still leave S stationary on the same paths, so invariance of $\mathcal{L}$ is sufficient but not necessary.

$$\frac{\partial \mathcal{L}}{\partial \dot{q}} \dot{q} - \mathcal{L} = \text{const.} = \mathcal{H},$$

which defines the *Hamiltonian*. Whenever the kinetic energy is quadratic in the velocities and the coordinates carry no explicit time dependence—the ordinary case—this evaluates to

$$\mathcal{H} \equiv T + V, \tag{10}$$

the total energy of the system.

3.3.2 Conservation of momentum

Spatial symmetry (homogeneity of space) leads to the law of conservation of momentum. If the Lagrangian does not depend explicitly on position,

$$\frac{\partial \mathcal{L}}{\partial \vec{r}} = 0,$$

then inserting this into [Eq. 8](#) leads to

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{\vec{r}}} \right) = 0.$$

With $\vec{p} \equiv \partial \mathcal{L} / \partial \dot{\vec{r}}$, we have

$$\frac{d\vec{p}}{dt} = 0 \Rightarrow \vec{p} = \text{const.},$$

so that the momentum is conserved.

For contrast, take a particle of mass m in a uniform gravitational field, for which

$$\mathcal{L} = \frac{1}{2} m \dot{\vec{r}}^2 - mgz.$$

This does depend on position, through z : moving the whole experiment upwards changes the potential energy, so space is not homogeneous in that direction. The vertical momentum is accordingly not conserved and the particle accelerates downwards. The horizontal coordinates x and y still do not appear, so p_x and p_y are conserved. A conservation law follows from a symmetry only in the directions in which that symmetry actually holds.

3.3.3 Conservation of angular momentum

Symmetry under rotation (isotropy of space) leads to the law of conservation of angular momentum. Turning the whole system through a small angle $\delta \vec{\phi}$ shifts every position by $\delta \vec{r} = \delta \vec{\phi} \times \vec{r}$ and every velocity by $\delta \dot{\vec{r}} = \delta \vec{\phi} \times \dot{\vec{r}}$. Requiring the Lagrangian not to notice,

$$\delta\mathcal{L} = \frac{\partial\mathcal{L}}{\partial\vec{r}} \cdot \delta\vec{r} + \frac{\partial\mathcal{L}}{\partial\dot{\vec{r}}} \cdot \delta\dot{\vec{r}} = 0,$$

and substituting Eq. 8 together with $\vec{p} \equiv \partial\mathcal{L}/\partial\dot{\vec{r}}$ collapses this to

$$\delta\vec{\phi} \cdot \frac{d}{dt} (\vec{r} \times \vec{p}) = 0.$$

The axis $\delta\vec{\phi}$ was arbitrary, so the bracket itself must be constant and the angular momentum is conserved,

$$\vec{L} \equiv \vec{r} \times \vec{p} = \text{const..}$$

In words: if turning the apparatus to face a different direction makes no difference to how it behaves, the combination $\vec{r} \times \vec{p}$ cannot change with time.

Watch Allan Adams on it.

4 Newton's laws of motion and the limits of their applicability; classical mechanics as a limiting form of quantum mechanics; non-relativistic classical mechanics as the limit of relativistic classical mechanics

4.1 Newton's laws of motion

Newton's laws of motion are three laws lying at the foundation of classical mechanics, specifying the relations between the motion of a body and the forces acting on it, which is why they are also called the laws of motion. In quantum mechanics they are for the most part inapplicable, and in relativistic mechanics they hold only within a limited range.

Newton's first law

In an inertial frame of reference (Section 1), if no force acts on a body, or if the acting forces balance one another, then the body remains at rest or moves in uniform rectilinear motion.

The first law assumes the existence of an inertial frame of reference but does not indicate how such a frame is to be sought, and it is therefore a postulate of the existence of an inertial frame of reference—*there exists a frame of reference in which a particle, not subject to interaction with its surroundings, remains at rest or moves along a straight line with constant velocity.*

Newton's second law

If an unbalanced force acts on a body, then the body moves with an acceleration directly proportional to the force and inversely proportional to the mass of the body.

$$\vec{F} = m\vec{a}. \quad (11)$$

Newton's third law

Interactions between bodies are always mutual. In an inertial frame of reference the forces of mutual interaction of two bodies have the same magnitude, opposite directions, and different points of application (each acts on a different body).

$$\vec{F}_{12} = -\vec{F}_{21} \quad (12)$$

In classical mechanics, Newton's third and second laws combined are equivalent with the law of conservation of momentum.

4.2 Limits of applicability of Newton's laws

Newton's laws of motion lose their direct validity in non-inertial frames, where it becomes necessary to introduce fictitious forces (sourceless, not subject to the 3rd law). Newtonian mechanics also becomes more complicated in more intricate co-ordinate systems. In such cases the Lagrangian or Hamiltonian formalisms, based on the principle of least action ([Section 7](#)), work better.

Quantum mechanics describes the behaviour of particles on small scale, where Newtonian mechanics fails. For example, the uncertainty principle ([Section 18](#)) tells us that we cannot simultaneously arbitrarily well know the position and the momentum of a particle, whereas classical mechanics rests *on their exact trajectories*.

The results of quantum mechanics in the limit of large quantum numbers should generally agree with the results of classical mechanics (Bohr's correspondence principle). In particular, assuming $\hbar \rightarrow 0$ may be regarded as a transition from a discrete to a continuous system. Some phenomena, however, have no classical limit at all: the stability of the atom and the discreteness of its spectrum ([Section 5.6](#)), tunnelling through a barrier ([Section 21](#)), and the exclusion of identical fermions from a common state ([Eq. 61](#)), which is what gives matter its volume and the periodic table its structure.

Moreover, Newton assumed that time is absolute, which was only countered 300 years later by Einstein and his theory of relativity, where the Galilean transformation is replaced by the Lorentz transformation ([Section 1](#)) and the two agree only to the extent that $\gamma \rightarrow 1$ for $v \ll c$ ([Eq. 3](#)). For this reason classical mechanics still remains a useful tool for describing non-relativistic phenomena. Again, certain effects are inherently relativistic even though they are directly observable: cosmic-ray muons reach sea level in far greater numbers than their lifetime would allow without time dilation, and the atomic clocks aboard GPS satellites have to be corrected for both special- and general-relativistic effects or the fix drifts by kilometres a day.

5 The postulates of quantum mechanics; measurement, state of a physical system

Quantum mechanics rests on a handful of *postulates*—statements that are not derived from anything deeper, but assumed, and then justified by the fact that everything they predict turns out to be correct. They answer three questions: what a state *is*, what a measurement *does*, and how a state *changes* in time.

5.1 The wave function

Classically, to know a particle is to know its position and momentum at every instant. Quantum mechanics replaces both by a single complex-valued function of position and time, the *wave function* $\Psi(x, t)$.

Postulate I — the state. *The state of a system is completely described by a wave function $\Psi(x, t)$. Everything that can be known about the system is contained in it.*

The wave function is not itself observable. Experiments return real numbers, and Ψ is complex, so it cannot be measured. What can be measured is its squared modulus $|\Psi|^2 = \Psi^* \Psi$, which is real and never negative.

Postulate II — the Born rule. $|\Psi(x, t)|^2$ is a probability density: the probability of finding the particle between a and b at time t is

$$P_{a \leq x \leq b}(t) = \int_a^b |\Psi(x, t)|^2 dx. \quad (13)$$

Because the particle must be somewhere, the wave function is *normalised*,

$$\int_{-\infty}^{\infty} |\Psi(x, t)|^2 dx = 1.$$

This forces two requirements that do real work later. Ψ must be *single-valued*, meaning one value of Ψ at each (x, t) , and it must die away fast enough at large $|x|$ for the integral to converge. Note also that multiplying Ψ by a constant phase $e^{i\alpha}$ changes no probability, so two states differing only by such a *global phase* are physically the same state.

5.2 Superposition

Postulate III — superposition. *Any sum of possible states is itself a possible state: if Ψ_1 and Ψ_2 are states, so is $c_1 \Psi_1 + c_2 \Psi_2$ with complex c_1, c_2 .*

If a system can be prepared in state A and in state B, it can be prepared in the combination ‘A + B’, realising both alternatives at once. The equation governing Ψ turns out to be linear (Eq. 18), so each term evolves on its own and the system stays a superposition at every later instant.

5.3 Why the wave function is complex: interference

A complex number whose squared modulus gives a probability is called a *probability amplitude*. It carries a magnitude *and* a phase, and the phase is what separates quantum from classical probability. Probabilities are positive and cannot cancel. Amplitudes can. This is forced on the theory rather than chosen for convenience: the i sitting on the left of Eq. 18 means Ψ cannot be a real function, and without a phase there would be nothing to cancel and no interference at all.

Two rules govern them. Amplitudes for steps taken *in succession* multiply. Amplitudes for *alternative* routes to the same outcome add, provided nothing in the experiment records which route was taken. The squared modulus is taken only at the very end.

In the double slit, let A_1 and A_2 be the amplitudes for a photon to reach a chosen point of the screen through slit 1 and slit 2. With both slits open and no record of which was used, the alternatives are *indistinguishable*, the amplitudes add, and

$$P = |A_1 + A_2|^2 = |A_1|^2 + |A_2|^2 + 2 \operatorname{Re}(A_1^* A_2).$$

The last term is the *interference term*. It depends on the relative phase of the two amplitudes, which shifts across the screen as the path difference changes, and it is what produces the fringes. Add a detector that records the slit and the alternatives become distinguishable, so it is now the probabilities that add, $P = |A_1|^2 + |A_2|^2$, with no interference and no fringes. Amplitudes add when the alternatives are not distinguished, probabilities add when they are. The information that would tell them apart is called *which-path information*, and merely having it available in the apparatus is enough to destroy the fringes.

5.4 Observables as operators

Position probabilities come straight out of Eq. 13. Every other measurable quantity has to be extracted from Ψ , and the tool for that is an *operator*: an instruction that takes one function and returns another. Position acts by multiplication and momentum by differentiation,

$$\hat{x} \Psi = x \Psi, \quad \hat{p} = -i\hbar \nabla, \quad \hat{p}_x = -i\hbar \frac{\partial}{\partial x}.$$

The momentum operator is read off from the de Broglie relation $p = \hbar k$ (Eq. 50): differentiating a plane wave e^{ikx} brings down a factor ik , and $ik \cdot (-i\hbar) = \hbar k$. The operator of the total energy, the *Hamiltonian*, is then built from it exactly as the classical $\mathcal{H} = T + V$ (Eq. 10) is,

$$\hat{H} = \frac{\hat{p}^2}{2m} + V(x) = -\frac{\hbar^2}{2m} \nabla^2 + V(x),$$

where

$$\nabla \equiv \begin{bmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \\ \frac{\partial}{\partial z} \end{bmatrix}. \quad (14)$$

Postulate IV — observables. *To every measurable quantity q there corresponds a Hermitian operator $\hat{Q}$ acting on the wave function. Hermitian means $\hat{Q} = \hat{Q}^\dagger$, where $\dagger$ denotes transposition combined with complex conjugation.*

The restriction is not arbitrary. Hermitian operators have real eigenvalues, as measured numbers must be, and their eigenfunctions belonging to different eigenvalues are orthogonal and together form a complete set. The possible outcomes are therefore mutually exclusive and exhaust the possibilities.

5.5 Eigenstates and measurement

An operator normally changes the function it acts on. The exceptional case is when it merely rescales it,

$$\hat{Q} \psi_n = q_n \psi_n, \quad (15)$$

where ψ_n is an *eigenfunction* of $\hat{Q}$ and the number q_n its *eigenvalue*. These are the states in which the observable has a definite value, and they are the exception rather than the rule: almost every state *is not* an eigenfunction of a given operator.

Because the eigenfunctions form a complete set, any state can be written as a sum of them,

$$\Psi = \sum_n c_n \psi_n, \quad c_n \in \mathbb{C}, \quad (16)$$

and because they are *orthonormal*, $\int \psi_n^* \psi_m dx = \delta_{nm}$ (the Kronecker delta, 1 when $n = m$ and 0 otherwise), each coefficient is recovered by $c_n = \int \psi_n^* \Psi dx$.

Postulate V — measurement. *The only possible result of measuring $\hat{Q}$ is one of its eigenvalues q_n , obtained with probability $|c_n|^2$. Immediately after a measurement returning q_n , the system is in the corresponding eigenstate ψ_n .*

The Born rule of [Eq. 13](#) is the special case of this for position. The final sentence is the *collapse* of the wave function, and it is the one step in the theory that is neither smooth nor reversible: a state carrying a whole range of possible answers becomes, abruptly, a single eigenfunction. Why this happens is unexplained and is called the *measurement problem*. If several eigenfunctions share one eigenvalue (a *degenerate* outcome), the measurement leaves behind only the part of the state built from those particular eigenfunctions, renormalised, rather than one definite eigenfunction.

Which eigenfunctions you expand in depends on what you intend to measure. The eigenfunctions of $\hat{H}$ are not those of $\hat{p}$. The state is one object, written down in

whichever set matches the apparatus, much like the same vector written in different bases.

Applying Eq. 15 to the energy gives the *time-independent Schrödinger equation*,

$$\boxed{\hat{H} \psi(x) = E \psi(x).} \quad (17)$$

Despite the name, this is not a wave equation—there is no time in it. It is an energy eigenvalue equation, and its solutions are the states of definite energy.

5.6 Where quantisation comes from

Quantisation is *not* a postulate. It falls out of Eq. 17 once the conditions on Ψ are imposed: the wave function must be finite, single-valued, normalisable and continuous, and its first derivative must be continuous as well wherever the potential is finite⁴.

When the particle is *bound*, meaning trapped in a potential well, these conditions can be met only for particular values of E , and the energy spectrum is discrete. An unbound particle is not boxed in by them and its spectrum is continuous. The mechanism is the one that gives a guitar string fixed at both ends its discrete set of frequencies. The difference, and it is not a small one, is that here it is the *energy* that is discrete, which classical physics never predicts.

- **Infinite square well** of width L : $E_n = n^2 \pi^2 \hbar^2 / 2mL^2$. Tighter confinement means larger spacing. The lowest state has $E_1 \neq 0$, the *zero-point energy*: a confined quantum particle can never be completely at rest.
- **Harmonic potential** $V \propto x^2$: evenly spaced levels $E_n = (n + 1/2) \hbar \omega$, with gap $\hbar \omega$. Any potential minimum looks harmonic for small displacements, so this is the generic case for vibrations of molecules and solids.
- **Hydrogen atom**: the confining agent is the electrostatic attraction of the nucleus, $V(r) = -e^2 / 4\pi \epsilon_0 r$, so the field of the nucleus is what does the confining. The bound states ($E < 0$) form the discrete series $E_n = -13.6 \text{ eV} / n^2$, crowding together as n grows because the well is shallower further out. Above $E = 0$ the electron is free and the spectrum is continuous.
- **Angular momentum** is quantised for a related but distinct reason: the wave function must return to itself after a full turn, which forces $L_z = m \hbar$ with integer m . This is confinement in angle rather than in radius.
- **Free particle**: confined by nothing, so neither its energy nor its momentum is quantised.

Light is also exchanged in quanta, $E = h\nu = \hbar \omega$. This does *not* follow from the postulates above, which describe a fixed set of particles and say nothing about the electromagnetic field. It is an experimental input, and it is recovered as a result only once the field itself is treated quantum-mechanically.

⁴Where the potential is infinite there is no tunnelling, the wave function is forced to zero at the boundary, and its derivative is allowed to jump there.

5.7 Time evolution

Postulate VI — time evolution. *Between measurements the state evolves according to the time-dependent Schrödinger equation,*

$$\boxed{i\hbar \frac{\partial \Psi}{\partial t} = \hat{H} \Psi.} \quad (18)$$

The Schrödinger equation is a guess. It is not derived from anything deeper, and the reasoning behind it runs backwards, from the answer to the equation. Start from a de Broglie wave, $\Psi \propto e^{i(kx - \omega t)}$, and from two relations already supplied by experiment, $E = \hbar\omega$ and $p = \hbar k$. What we want to reproduce is the ordinary classical statement $E = p^2/2m + V$, so look at the powers in it: energy appears once, momentum appears squared. We therefore need something that pulls *one* factor of ω out of the wave, and something that pulls *two* factors of k out of it.

Differentiation does precisely that, because each derivative of an exponential brings its exponent down as a multiplying factor. Dressed with the right constants,

$$\begin{aligned} i\hbar \frac{\partial \Psi}{\partial t} &= i\hbar (-i\omega) \Psi = \hbar\omega \Psi = E \Psi, \\ -\frac{\hbar^2}{2m} \frac{\partial^2 \Psi}{\partial x^2} &= -\frac{\hbar^2}{2m} (ik)^2 \Psi = \frac{\hbar^2 k^2}{2m} \Psi = \frac{p^2}{2m} \Psi. \end{aligned}$$

This is the point of the exercise. Each combination hands back *the same wave multiplied by the number we were after*, which is exactly the eigenvalue pattern of [Eq. 15](#): differentiation is the machinery that extracts a physical number from a wave. Substituting these two expressions for E and $p^2/2m$ in the classical energy statement gives [Eq. 18](#). The equation is, in effect, reverse-engineered—the simplest linear equation whose plane-wave solutions carry the right energy and the right momentum. It is not a fundamental law in the way Maxwell’s equations are. It is kept because everything it predicts turns out to be right.

Because the equation is *first* order in time, unlike a classical wave equation, the state at one instant fixes it at every other. The evolution keeps the total probability equal to 1 and is fully reversible, and evolution of that kind is called *unitary*. It stands in sharp contrast to the abrupt, irreversible collapse of the measurement postulate, and the tension between the two *is* the measurement problem.

If the Hamiltonian does not itself depend on time, [Eq. 18](#) separates into a space part and a time part. The space part is [Eq. 17](#), and the full solution differs from it only by a rotating phase,

$$\Psi(x, t) = \psi(x) e^{-\frac{i}{\hbar} E t}. \quad (19)$$

This fixes the convention used throughout the note: capital Ψ or Φ is the full, time-dependent wave function, lower-case ψ or ϕ the time-independent part alone. Such a

state is called *stationary*, because the phase cancels in every squared modulus and so no probability changes with time. The phase turns at angular frequency $\omega = E/\hbar$, so $E = \hbar\omega$ appears here as a property of the state, not as a statement about photons.

On its own a rotating phase does nothing. In a superposition it does a great deal. Take a state built from two stationary states of different energy,

$$\Psi = c_1\psi_1e^{-\frac{i}{\hbar}E_1t} + c_2\psi_2e^{-\frac{i}{\hbar}E_2t}.$$

The two phases turn at different rates, so the phase *difference* between the terms grows steadily, and the interference term in $|\Psi|^2$ oscillates at the frequency $\omega_{21} = (E_2 - E_1)/\hbar$. The probability density sloshes back and forth at that frequency instead of standing still, and for an electron in an atom ω_{21} is precisely the frequency of the light emitted or absorbed on that transition. A *global* phase changes nothing, a *relative* phase changes everything.

5.8 States as vectors: Dirac notation

Equation 16 fixes the state entirely by the list of coefficients c_n , and a list of numbers is a vector. This is the starting point of Heisenberg's *matrix mechanics*: wave functions are vectors, and the eigenfunctions are the basis vectors of the space they live in.

That space is a *Hilbert space* $\mathcal{H}$: a complex vector space equipped with a scalar product, of finite or infinite dimension. The abstract vector is written $|\Psi\rangle$ and called a *ket*, so that **Eq. 16** becomes $|\Psi\rangle = \sum_n c_n |\psi_n\rangle$. In a basis, $|\Psi\rangle$ is a column of complex numbers, and the *bra* $\langle\Psi| = |\Psi\rangle^\dagger$ is the corresponding row with every entry conjugated.⁵

The scalar product is inherited from the overlap integral of two wave functions,

$$\langle\Phi|\Psi\rangle = \int \Phi^*(x) \Psi(x) dx = \sum_n b_n^* c_n \in \mathbb{C}, \quad (20)$$

where b_n and c_n are the expansion coefficients of the two states. The conjugation makes it Hermitian rather than symmetric, $\langle\Phi|\Psi\rangle = \langle\Psi|\Phi\rangle^*$, so that $\langle\Psi|\Psi\rangle \geq 0$ is real and its square root is the length of the vector. Orthonormality reads $\langle\psi_n|\psi_m\rangle = \delta_{nm}$ with the Kronecker delta as before, the coefficients are recovered as $c_n = \langle\psi_n|\Psi\rangle$, and the measurement postulate takes its compact form,

$$P_n = |\langle\psi_n|\Psi\rangle|^2. \quad (21)$$

⁵The two possible orders of multiplication give quite different objects. Row times column, $\langle\Phi|\Psi\rangle = \begin{bmatrix} b_1^* & b_2^* \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = b_1^* c_1 + b_2^* c_2$, is 1×2 times 2×1 , so the result is 1×1 : a single number.

Column times row, $|\Psi\rangle\langle\Phi| = \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} \begin{bmatrix} b_1^* & b_2^* \end{bmatrix} = \begin{bmatrix} c_1 b_1^* & c_1 b_2^* \\ c_2 b_1^* & c_2 b_2^* \end{bmatrix}$, is 2×1 times 1×2 , so the result is 2×2 : a matrix, that is, an operator.

In this language an operator on an m -dimensional space is an $m \times m$ matrix, Hermiticity is the statement that the matrix equals its own conjugate transpose, and Eq. 15 is an ordinary matrix eigenvalue problem.

5.9 The simplest example: spin

Some degrees of freedom have no wave function in position at all, and for these the vector language is the only one available. Electron spin is the smallest case, a two-dimensional Hilbert space (spin itself is treated in Section 6). Take the states $|\uparrow\rangle$ and $|\downarrow\rangle$, spin up and down along z , as the basis:

$$|\uparrow\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad |\downarrow\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad \hat{S}_z = \frac{\hbar}{2} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}.$$

Multiplying out gives $\hat{S}_z |\uparrow\rangle = +\hbar/2 |\uparrow\rangle$ and $\hat{S}_z |\downarrow\rangle = -\hbar/2 |\downarrow\rangle$, so the basis states are eigenvectors and the only possible results are $\pm\hbar/2$. Now take the superposition

$$|\Psi\rangle = \frac{1}{\sqrt{2}} (|\uparrow\rangle + |\downarrow\rangle) = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

The amplitude for spin up is $\langle\uparrow|\Psi\rangle = 1/\sqrt{2}$, so by Eq. 21 the probability is $1/2$, and likewise for spin down. Acting with $\hat{S}_z$ gives $\hbar/2 \cdot 1/\sqrt{2} (|\uparrow\rangle - |\downarrow\rangle)$, which is not a multiple of $|\Psi\rangle$: the operator has genuinely rotated the vector, as it does for any state that is not one of its eigenvectors. The same $|\Psi\rangle$ is nonetheless an exact eigenvector of $\hat{S}_x = \hbar/2 \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ with eigenvalue $+\hbar/2$. Having a sharp x -component costs it all knowledge of the z -component, which is the uncertainty principle (Section 18) in its simplest form.

5.10 Mean values

Ordinary probability theory defines a mean in two equivalent ways. For a quantity with discrete possible values q_n occurring with probabilities P_n , the mean is $\sum_n q_n P_n$. For a continuous variable with normalised probability density $P(x)$, it is $\langle x \rangle = \int x P(x) dx$.

Quantum mechanics uses exactly these, with $P_n = |c_n|^2$. A single measurement returns one eigenvalue. Repeating it on many identically prepared systems gives a distribution whose mean is

$$\langle \hat{Q} \rangle = \sum_n q_n |c_n|^2 = \langle \Psi | \hat{Q} | \Psi \rangle = \int \Psi^*(x) \hat{Q} \Psi(x) dx.$$

The forms agree because the operator can be rebuilt from its own outcomes and eigenstates,

$$\hat{Q} = \sum_n q_n |\psi_n\rangle \langle \psi_n|, \quad (22)$$

where each $|\psi_n\rangle\langle\psi_n|$ is a *projection operator*, a matrix that discards everything in a state except its $|\psi_n\rangle$ component. (Equation 22 as written assumes a discrete, non-degenerate spectrum.) The mean is generally *not* one of the possible outcomes: for the spin state above $\langle\hat{S}_z\rangle = 0$, a value never seen in any single run.

5.11 Systems with several degrees of freedom

If a system has more than one degree of freedom (a photon carries both a position and a polarisation), each degree of freedom has its own Hilbert space $\mathcal{H}_i$.

Postulate VII — composite systems. *The Hilbert space of a composite system is the tensor product of the spaces of its parts, $\mathcal{H} = \bigotimes_i \mathcal{H}_i$.*⁶

A state that cannot be written as a single such pair is called *entangled*. Take two spins in

$$|\Psi\rangle = \frac{1}{\sqrt{2}} (|\uparrow\uparrow\rangle + |\downarrow\downarrow\rangle),$$

where $|\uparrow\downarrow\rangle$ means the first spin up and the second down. No choice of a state for the first spin and a state for the second reproduces this, so neither particle has a state of its own—only the pair does.

It follows from what is already above rather than from anything new. By the superposition postulate the pair may sit in a combination of the two joint alternatives, and by the measurement postulate the collapse applies to the pair as a whole. Measuring the first spin gives up or down with probability 1/2 each, and whichever comes out, the state collapses to a single term and the second spin is then certain to match, however far apart the two have been carried. This is why the tensor product is more than bookkeeping: most states in the product space are not products at all.

6 Angular momentum and the phenomenon of precession — classical and quantum description

The classical angular momentum of a system is defined as

$$\vec{L} \equiv \sum_i \vec{r}_i \times \vec{p}_i, \quad (23)$$

⁶The tensor product $\mathcal{H}_1 \otimes \mathcal{H}_2$ is the space whose basis vectors are all the pairs formed by taking one basis vector from $\mathcal{H}_1$ and one from $\mathcal{H}_2$. Its dimensions therefore multiply rather than add: a two-state spin combined with a three-state degree of freedom gives a six-dimensional space, not a five-dimensional one.

where $\vec{r}_i$ is the position and $\vec{p}_i$ the momentum of the i -th element of the system. It is the quantity conserved when space is isotropic (Section 3).

In quantum mechanics the same expression is promoted to an operator (postulate IV), the *orbital* angular momentum

$$\hat{L} = -i\hbar (\vec{r} \times \nabla). \quad (24)$$

A particle also carries *spin*, an angular momentum that is not associated with any motion through space. It transforms under rotations exactly as $\hat{L}$ does, but it has no classical counterpart and is not a picture of the particle turning about itself. Like $\hat{L}$ it is a vector of three operators, one observable per axis,

$$\hat{S} \equiv \begin{bmatrix} \hat{S}_x \\ \hat{S}_y \\ \hat{S}_z \end{bmatrix}, \quad (25)$$

and for an electron each component is a 2×2 matrix. This is the operator already used as the worked example in Section 5, where $\hat{S}_z$ was written out, its eigenvectors $|\uparrow\rangle$ and $|\downarrow\rangle$ identified, and a measurement on a superposition of them followed through. That discussion is the concrete content of Eq. 25 and is not repeated here.

The total angular momentum is the sum of the two,

$$\hat{J} = \hat{L} + \hat{S}.$$

6.1 Quantisation and quantum numbers

Only two things about an angular momentum can be known at once: the magnitude of its square, and one chosen component, conventionally the one along z . The other two components then have no definite value. Both of the knowable quantities are quantised,

$$\hat{L}^2 |l, m\rangle = \hbar^2 l(l+1) |l, m\rangle, \quad \hat{L}_z |l, m\rangle = \hbar m |l, m\rangle,$$

with $l = 0, 1, 2, \dots$ and $m = -l, -l+1, \dots, +l$, so an angular momentum of a given magnitude has $2l+1$ allowed orientations. The same pattern holds for $\hat{S}$ with the pair (s, m_s) and for $\hat{J}$ with (j, m_j) , except that s and j are allowed to be half-integer. An electron always has $s = 1/2$ and therefore $m_s = \pm 1/2$, which is where its two states come from.

For one electron in a hydrogen atom the numbers divide the work between them.

- **Principal**, $n = 1, 2, 3, \dots$ Fixes the energy, $E_n = -13.6 \text{ eV}/n^2$ (Section 5.6), and roughly the size of the orbital.
- **Orbital**, $l = 0, \dots, n-1$ Fixes how much orbital angular momentum the electron has, $|\vec{L}| = \hbar\sqrt{l(l+1)}$, and hence the shape of the orbital. These are the labels s, p, d, f.

Not sure if Hydrogen here fits but some section should discuss quantum numbers and fine/hyperfine structure. Avoid repetition. Keep it in one place and refer to it from elsewhere.

- **Magnetic**, $m = -l, \dots, +l$ Fixes the tilt of that angular momentum, its projection on a chosen axis. Nothing distinguishes the $2l+1$ values until a magnetic field selects a direction, at which point the level splits into $2l+1$ (the Zeeman effect).
- **Spin**, $m_s = \pm 1/2$ The projection of the spin, doubling every count above.

Since $\vec{L}$ and $\vec{S}$ belong to the same electron, they combine into $\vec{J}$, and j then runs from $|l-s|$ to $l+s$ in integer steps—for $l=1$ and $s=1/2$, either $j=1/2$ or $j=3/2$. The two combinations differ slightly in energy because the electron's magnetic moment sits in the magnetic field produced by its own orbital motion, and that splitting is the *fine structure* of the spectrum.

6.2 Classical precession

Precession is the slow circular drift of the axis of a spinning object. Instead of pointing in a fixed direction, the spin axis itself sweeps around, tracing out the surface of a cone. The familiar example is a spinning toy top: as it spins, its axis does not simply fall over under gravity but slowly circles around the vertical.

The angular momentum $\vec{L}$ of the spinning body points along its spin axis, and a torque changes it at the rate $d\vec{L}/dt = \vec{\tau}$. If the torque is perpendicular to $\vec{L}$, then so is the increment it adds, so it turns $\vec{L}$ without lengthening it and the axis sweeps out a cone. For the toy top, gravity acting at the centre of mass supplies the torque, and unless the axis stands exactly vertical that torque is horizontal and therefore perpendicular to $\vec{L}$. It seems counterintuitive that the spin axis does not tilt in the direction of the force but it *should not*—it will tilt exactly in the direction of the supplied torque of the force.

The precession frequency is given by

$$\omega_p = \frac{\tau}{I_s \omega_s \sin(\theta)}$$

where τ is the torque, I_s the moment of inertia, ω_s the spin angular velocity, and θ the angle between the spin axis and the vertical axis about which it precesses.

6.3 Quantum precession

A particle such as an electron carries a magnetic moment $\vec{\mu}$ associated with its spin angular momentum $\vec{S}$ —the two are proportional, $\vec{\mu} = \gamma \vec{S}$ with γ the *gyromagnetic ratio*, and so lie along the same axis. Placed in an external magnetic field $\vec{B}$, the moment feels a torque

$$\vec{\tau} = \vec{\mu} \times \vec{B},$$

which is perpendicular to $\vec{S}$ and therefore, exactly as in the classical case, makes $\vec{S}$ sweep around the direction of $\vec{B}$ instead of aligning with it. This is Larmor precession, the direct analogue of gravity making a toy top precess, with the spin angular momentum $\vec{S}$ playing the role of the top's angular momentum $\vec{L}$.

The rate of this precession, the *Larmor frequency* $\omega_L = \gamma B$, depends only on the field strength and not at all on how far $\vec{S}$ is tilted away from $\vec{B}$, because the two factors of $\sin \theta$ that enter the torque and the cone radius cancel. Every spin in a sample therefore turns at the same rate whatever its orientation, which is what makes the effect usable for magnetic resonance.

watch youtube
about it

7 Variational principles of physics; examples of equations that can be derived from them

7.1 What a variational principle is

There are two complementary ways to describe how a system moves. Newton's is *local*: at each instant the force fixes the acceleration, the acceleration updates the velocity, and the trajectory is assembled step by step. It requires every force to be known as a vector and resolved along fixed axes, forces of constraint included. A variational principle takes the opposite, *global* view. It considers every path the system could conceivably follow between a fixed start point and a fixed end point, attaches a number to each, and asserts that the path actually taken is the one at which that number is stationary. It needs only the two scalars T and V , in whatever coordinates suit the problem, and forces of constraint never appear.

That number is the *action* S (Eq. 7), which can be evaluated for any candidate path and not only for the true one. A quantity that takes in a whole function and returns a single number is called a *functional*, as opposed to an ordinary function, which takes in a number and returns a number.

The *principle of least action* says that the path followed is the one for which the action is *stationary*: deform the path slightly in any way, holding the endpoints fixed, and to first order the action does not change, written $\delta S = 0$. This is the same condition as $f'(x) = 0$ at the minimum of an ordinary function, with the single variable x replaced by an entire path. Only in the simplest cases is the stationary path a true minimum, so 'least' is something of a historical misnomer—a maximum or a saddle satisfies the condition equally well.

The idea began in optics as *Fermat's principle*: light travels between two points along the route taking the least time, from which the laws of reflection and refraction follow (Eq. 46). The machinery came from a different kind of problem, the *brachistochrone* (Johann Bernoulli, 1696): of all ramps joining a point to a lower one, find the shape down which a bead released from rest slides in the shortest time. The answer is a cycloid, the curve traced by a point on the rim of a rolling wheel. That is an optimisation posed by the experimenter rather than a law of nature—the bead does not choose its own ramp—so it is not itself an instance of the principle of least

action. What the two share is the mathematics: in each, a whole curve is singled out by requiring an accumulated integral to be stationary. Problems of that shape are what the *calculus of variations* of Euler and Lagrange was built for, and it is in that language that the principle of least action is written.

Check the example online. Maybe skip this historical nuance here.

7.2 Equations that follow from it

- **Euler–Lagrange equation.** Imposing $\delta S = 0$ on the mechanical action—the Lagrangian (Eq. 6) integrated over time—converts the loose instruction ‘choose the best path’ into a definite differential equation the path must satisfy at every instant (Eq. 8). For a single particle it reproduces Newton’s second law (Eq. 11), but now in whatever coordinates are convenient. These *generalised coordinates* q_i are any set of numbers that fixes the configuration of the system, such as an angle or a distance along a track, and the freedom to choose them is what makes the method far easier than Newton’s for constrained or awkwardly shaped systems.
- **Hamilton’s equations.** A standard change of variables (a *Legendre transformation*) trades the velocities $\dot{q}_i$ for momenta and repackages the same content as the *Hamiltonian* formulation, built on the total energy (Eq. 10). This is the usual starting point for statistical physics and for quantum mechanics.
- **Hamilton–Jacobi equation.** Treating the action as a function of the point at which the path ends turns the problem into a single equation for the action itself (here called Hamilton’s principal function), in which every momentum is replaced by a derivative of the action, $p_i = \partial S / \partial q_i$. Solving that one equation is an alternative route to the motion, and its form is strikingly close to the Schrödinger equation (Eq. 18), which is why it is regarded as the classical doorway to wave mechanics.
- **Feynman’s path integral.** The fully quantum version of the whole scheme. Instead of selecting one path it gives every path a contribution $e^{iS/\hbar}$ and adds them all together. Where S is large compared with $\hbar$, neighbouring paths arrive with wildly different phases and cancel, except close to the stationary path where the phase barely varies—so the classical principle is recovered as a limit rather than assumed.
- **Maxwell’s equations.** The recipe is not restricted to mechanics: write down an action for the system and demand that it be stationary. The action of the electromagnetic field, together with its coupling to charges, yields Maxwell’s equations (Eq. 30).
- **Einstein’s field equations.** The same demand applied to the Einstein–Hilbert action, whose integrand measures the curvature of spacetime, yields the field equations of general relativity. That one rule generates the governing equations of mechanics, electromagnetism and gravitation alike is what makes the variational viewpoint one of the unifying ideas of physics.

8 The laws of thermodynamics

Thermal equilibrium implies there is no net flow of thermal energy between two systems that can exchange heat. A system is said to be in thermal equilibrium with itself if the temperature within the system is spatially uniform and temporally constant.

8.1 The zeroth law of thermodynamics

If systems A and B, capable of exchanging heat with one another, are in thermal equilibrium with each other, and the same is true of systems B and C, then systems A and C are also in thermal equilibrium with each other.

8.2 The first law of thermodynamics

The change in the internal energy of a system is equal to the sum of the heat supplied and the *work done on the system*.

$$\Delta U = Q + W, \quad (26)$$

or in a differential form:

$$dU = dQ - p dV, \quad (27)$$

where dV is the change in volume of the system (positive when the system expands—it does work and loses internal energy, $W < 0$).

8.3 The second law of thermodynamics

- Clausius statement: *There is no thermodynamic process whose sole result would be the absorption of heat from a reservoir at a lower temperature and its transfer to a reservoir at a higher temperature.*
- Kelvin statement: *No process is possible whose sole effect would be the absorption of a certain quantity of heat from a reservoir and its conversion into an equivalent amount of work.*
- Planck statement: *It is impossible to construct an engine which will work in a complete cycle, and produce no effect except the production of work and cooling of a heat reservoir.*

These statements are made quantitative by the *Clausius inequality*, which holds for any process:

$$dS \geq \frac{\delta Q}{T}. \quad (28)$$

Here S is the *entropy* and T the temperature at which the heat δQ crosses the boundary. Entropy is a *function of state*, meaning its change between two equilibrium

states depends only on those two states and not on the path taken between them, and that is what makes the left-hand side usable even though no instrument reads entropy directly. To obtain ΔS for a real process, however violent, one connects the same two end states by *any* convenient reversible path and integrates $\delta Q/T$ along that one instead, which is ordinary calorimetry. Equation 28 then measures the real process against its reversible ideal: the entropy actually gained is never less than the heat transfer accounts for, and the excess is entropy generated by the irreversibility itself. This is what caps the efficiency of every heat engine and fixes which way a process is allowed to run. The equality holds only for reversible processes. For an isolated system ($\delta Q = 0$) it reduces to

$$dS \geq 0,$$

so spontaneous processes drive the system towards higher entropy and equilibrium is the state of maximum entropy. This macroscopic, thermodynamic entropy coincides with the microscopic count of accessible microstates (Eq. 62), which is why a system left to itself drifts towards the macrostate realised by the overwhelmingly largest number of microstates.

A process is *reversible* if it can be run backwards through the same sequence of equilibrium states leaving no net change in either the system or its surroundings—an idealisation that requires it to be infinitely slow (quasi-static) and free of friction or other dissipation. Every real process is *irreversible*: it generates entropy, so the inequality in Eq. 28 is then strict.

8.4 The third law of thermodynamics

The Nernst heat theorem: *It is not possible, by means of a finite number of steps, to attain the temperature of absolute zero (zero kelvin) if we take a non-zero absolute temperature as our starting point.*

9 Mechanical properties of material media: liquids, elastic media, mechanical waves

Liquids

Liquids are substances intermediate between a gas and a solid, having an easily variable shape but little compressibility. The properties of a liquid follow from the behaviour of its molecules:

- they have complete freedom of movement within the volume occupied by the liquid

- *intermolecular interactions* act between them—the residual electrostatic attractions between whole neutral molecules—and within the bulk of the liquid these pull in every direction equally and cancel
- the intermolecular interactions do not cancel at the interface between the liquid and another phase, as a result of which the phenomenon of surface tension occurs.

Elastic media

An elastic medium is one in which a deformation gives rise to internal stresses that *oppose* it, so that the body returns to its original shape once the external force is removed. *Elastic* in the strict sense requires only that the deformation be fully recovered, not that the restoring force be proportional to it. Proportionality—the linear regime $F = -kx$, in which one may speak of an elastic modulus—is an extra assumption, though a safe one for small deformations: a system in stable equilibrium sits at a minimum of its potential energy, any smooth minimum looks like a parabola close up, and a parabolic potential gives a force linear in the displacement. Small disturbances therefore meet a linear elastic response whatever the medium does under large ones.

Solids stay elastic over a wide range of stresses. Fluids—liquids and gases alike—always resist a change of *volume*, so they possess bulk elasticity, but a fluid at rest offers no resistance whatever to a change of *shape*, having no static shear elasticity, which is precisely what allows it to flow. Compressing the air in a cylinder closed by a piston raises its pressure and it pushes back: that is bulk elasticity, and it is all that a mechanical wave requires.⁷ A fluid in the open shows an apparent *shape* elasticity only under fast forcing, when it has no time to flow out of the way.

Mechanical waves

Bulk elasticity is what carries a mechanical wave. Displace a small region of an elastic medium from its equilibrium position and it oscillates about that position, and because the medium is elastic the oscillation is passed on to neighbouring regions, so the disturbance travels through the medium as a whole. The speed of propagation is set by the medium alone, going roughly as the square root of its stiffness divided by its density. Waves are classified by the direction in which the medium oscillates relative to the direction of travel, which is also the direction in which energy is carried.

A wave is *longitudinal* when the medium oscillates parallel to the direction of propagation. Sound in air is the standard example, as is a spring alternately compressed and extended. A gas or a liquid can carry only this kind, since it has no shear elasticity to resist a sideways displacement.

A wave is *transverse* when the medium oscillates perpendicular to the direction

⁷Air in a balloon is the more familiar illustration but a worse one, since the rubber skin is itself elastic and contributes a restoring force of its own, which obscures where the elasticity is coming from.

of propagation. Examples are a plucked string and a vibrating ruler. Transverse mechanical waves need shear elasticity and so travel through solids only, which is how seismology distinguishes the Earth's liquid outer core from the solid mantle.

10 Flow of an incompressible liquid; Bernoulli's principle

Let us begin by defining the assumptions of the problem. It is an example from the field of ideal-fluid hydrodynamics, and therefore adopts certain simplifications:

- the fluid is incompressible, that is, it does not change its volume under pressure, which for liquids is usually a very good approximation
- friction arising from the viscosity of the liquid is negligible
- the flow of the liquid is steady and irrotational: at any fixed point in the pipe the velocity, density and so on do not change with time, and the fluid elements are carried along without spinning about themselves

Applying the principle of conservation of energy to the flow problem, one obtains Bernoulli's principle, which is usually written as:

$$\frac{v^2}{2} + gh + \frac{p}{\rho} = \text{const.} \quad (29)$$

where we have, in turn, the kinetic energy, the potential energy and the energy due to pressure, each per unit mass of fluid. The constant keeps the same value all along a *streamline*, meaning the path traced out by a small parcel of fluid as it is carried by the flow, and because the flow was assumed irrotational the same constant in fact serves the whole flow. The relation permits the description of flow through a pipe that changes its cross-section or its height, and it explains the so-called hydrodynamic paradox. Where a pipe narrows, incompressibility forces the liquid to flow faster, since the same volume must pass each second through a smaller area, so by Eq. 29 the larger $v^2/2$ must be offset by a smaller pressure p . The counter-intuitive result is that the pressure is *lower* where the pipe is narrow and the flow is fast.

11 Maxwell's equations, the scalar and vector potentials of the electromagnetic field, the equations of the electromagnetic field in vacuum and in material media

11.1 Maxwell's equations — general form

$$\begin{aligned}
 \text{Gauss's law for } \vec{E} : \quad \nabla \cdot \vec{E} &= \frac{\rho_f + \rho_b}{\epsilon_0} \\
 \text{Gauss's law for } \vec{B} : \quad \nabla \cdot \vec{B} &= 0 \\
 \text{Faraday's law : } \quad \nabla \times \vec{E} &= -\frac{\partial \vec{B}}{\partial t} \\
 \text{Ampère–Maxwell law : } \quad \nabla \times \vec{B} &= \mu_0 \left(\vec{J}_f + \vec{J}_b + \epsilon_0 \frac{\partial \vec{E}}{\partial t} \right),
 \end{aligned} \tag{30}$$

where ρ is a charge density, $\vec{J}$ a current density, $\vec{E}$ the electric field and $\vec{B}$ the magnetic induction field. Charges and currents are of two kinds. *Free* ones (ρ_f , $\vec{J}_f$) are the ones we put there and control, such as the conduction electrons in a wire or the charge deposited on a plate. *Bound* ones (ρ_b , $\vec{J}_b$) are locked inside the atoms of a material and can only shift or circulate slightly without leaving them, and it is the applied field itself that induces them. The fields make no such distinction—only the total charge and current act as sources—but it is worth making, because the bound part is a response of the material and can be absorbed into redefined fields (Section 11.3).

The nabla operator ∇ (Eq. 14) can be used to mean different operations, giving:

$$\begin{aligned}
 \nabla \cdot \vec{A} &\text{— divergence of a vector field} \\
 \nabla \times \vec{A} &\text{— curl of a vector field} \\
 \nabla f &\text{— gradient of a scalar field}
 \end{aligned}$$

Divergence measures how strongly a given point acts as a source or a sink of a vector field, curl measures the circulation of the field around a point, and the gradient points in the direction in which a scalar field increases fastest. Two integral theorems turn these statements about single points into statements about whole regions. The *divergence theorem*, $\int_V \nabla \cdot \vec{A} dV = \oint_{\partial V} \vec{A} \cdot d\vec{S}$, says that the divergence added up over a volume equals the flux out through the surface enclosing it. *Stokes' theorem*, $\int_S (\nabla \times \vec{A}) \cdot d\vec{S} = \oint_{\partial S} \vec{A} \cdot d\vec{l}$, says that the curl added up over a surface equals the circulation of the field around the loop bounding it. Applied to Eq. 30 they give each equation its plain reading:

- **Gauss's law for $\vec{E}$.** The electric flux out through any closed surface is fixed by the charge enclosed and by nothing else. Field lines start on positive charge and end on negative charge.
- **Gauss's law for $\vec{B}$.** The magnetic flux out through any closed surface is always zero, so magnetic field lines have no ends and close on themselves.

Check Grit-thits on bound currents

Theorems in their own lines + look up one-note notes about them and possibly restate.

There is no magnetic charge for them to start from—no magnetic monopole has ever been observed.

- **Faraday’s law.** The circulation of $\vec{E}$ around a loop equals minus the rate of change of the magnetic flux threading it, so a changing magnetic field drives a current in a wire. This is the generator and the transformer. The minus sign is Lenz’s law: the induced current opposes the change that produced it, as it must if energy is to be conserved.
- **Ampère–Maxwell law.** The circulation of $\vec{B}$ around a loop is fixed by the current passing through it together with the rate of change of the electric flux through it. The first term is the electromagnet. The second is Maxwell’s own addition, and it is what allows the two fields to sustain each other and travel as light through empty space (Eq. 32).

restate in more familiar way for me (give rise to...)

Fields act on charges through the Lorentz force:

$$\vec{F} = q(\vec{E} + \vec{v} \times \vec{B}) \quad (31)$$

11.2 Maxwell’s equations in vacuum

In vacuum there are no charges or currents at all, free or bound, so $\rho = 0$ and $\vec{J} = 0$, and Maxwell’s equations reduce to their source-free form:

$$\begin{aligned} \nabla \cdot \vec{E} &= 0 \\ \nabla \cdot \vec{B} &= 0 \\ \nabla \times \vec{E} &= -\frac{\partial \vec{B}}{\partial t} \\ \nabla \times \vec{B} &= \mu_0 \varepsilon_0 \frac{\partial \vec{E}}{\partial t} \end{aligned} \quad (32)$$

11.3 Maxwell’s equations in matter

An electric field pulls the positive and negative charge of each atom or molecule slightly apart, giving it an electric dipole moment $\vec{p} = q\vec{d}$, where $\vec{d}$ points from the negative charge to the positive one. Averaged per unit volume this is the polarisation vector

$$\vec{P} = \chi \varepsilon_0 \vec{E}, \quad (33)$$

where χ — electric susceptibility of a material. Polarisation moves bound charge only, and

$$\nabla \cdot \vec{P} = -\rho_b.$$

The minus sign records which charge is left behind. Since $\vec{P}$ points from the negative end of each dipole to the positive end, a region out of which $\vec{P}$ spreads (positive

divergence) is one from which positive bound charge has been carried away, so what remains there is negative.

From Gauss's law for the electric field, we then have

$$\begin{aligned}\nabla \cdot (\varepsilon_0 \vec{E}) &= \rho_f + \rho_b \\ \nabla \cdot (\varepsilon_0 \vec{E} + \vec{P}) &= \rho_f, \\ \nabla \cdot \vec{D} &= \rho_f,\end{aligned}$$

where $\vec{D}$ — the electric displacement field,

$$\vec{D} \equiv \varepsilon_0 \vec{E} + \vec{P}. \quad (34)$$

Adding $\vec{P}$ cancels the bound charge contributions, so $\vec{D}$ is the field whose sources are the free charges alone—the ones we control. $\vec{P}$ points in the opposite direction to the field it is producing, so it subtracts from $\vec{E}$.

Matter responds magnetically as well. Every atom carries a small circulating current and so a magnetic dipole moment, and the average moment per unit volume is the *magnetisation* $\vec{M}$. Inside a uniformly magnetised body, neighbouring loops run in opposite directions where they meet and cancel, so a net current survives only at the surface, or anywhere $\vec{M}$ varies from place to place. Added up, what is left is $\nabla \times \vec{M}$. A polarisation that changes in time is on top of that a genuine motion of bound charge, hence a current $\partial \vec{P} / \partial t$. The two together make the bound current

$$\vec{J}_b = \nabla \times \vec{M} + \frac{\partial \vec{P}}{\partial t}.$$

From the Ampère–Maxwell law, we then have

$$\begin{aligned}\nabla \times \vec{B} &= \mu_0 \left(\vec{J}_f + \vec{J}_b + \varepsilon_0 \frac{\partial \vec{E}}{\partial t} \right) \\ &= \mu_0 \left(\vec{J}_f + \nabla \times \vec{M} + \frac{\partial \vec{P}}{\partial t} + \varepsilon_0 \frac{\partial \vec{E}}{\partial t} \right) \\ \nabla \times \vec{H} &= \vec{J}_f + \frac{\partial \vec{D}}{\partial t},\end{aligned}$$

where $\vec{H}$ — the magnetic field intensity,

$$\vec{H} \equiv \frac{\vec{B}}{\mu_0} - \vec{M}. \quad (35)$$

The minus here does the same job as the plus in [Eq. 34](#). Moving $\nabla \times \vec{M}$ to the left-hand side of the equation above is what subtracts $\vec{M}$, and it removes the bound magnetisation current from the curl, leaving $\vec{H}$ sourced by free currents alone. Both

$\vec{D}$ and $\vec{H}$ are therefore devices for hiding the material's own response inside a re-defined field, so that only the sources we control appear explicitly.

Maxwell's equation in matter may then finally be rewritten as:

$$\begin{aligned}\nabla \cdot \vec{D} &= \rho_f \\ \nabla \cdot \vec{B} &= 0 \\ \nabla \times \vec{E} &= -\frac{\partial \vec{B}}{\partial t} \\ \nabla \times \vec{H} &= \vec{J}_f + \frac{\partial \vec{D}}{\partial t}\end{aligned}\tag{36}$$

11.4 The scalar and vector potentials

Two of Maxwell's equations carry no sources at all, and they can be satisfied identically—made true by construction—once the fields are written in terms of potentials.

Since $\nabla \cdot \vec{B} = 0$ everywhere, and since the divergence of any curl vanishes automatically, $\vec{B}$ may always be written as the curl of some vector field,

$$\vec{B} = \nabla \times \vec{A},\tag{37}$$

called the *magnetic vector potential*. Substituting this into Faraday's law gives $\nabla \times (\vec{E} + \partial \vec{A} / \partial t) = 0$, and a field whose curl vanishes is the gradient of something, so

$$\vec{E} = -\nabla \varphi - \frac{\partial \vec{A}}{\partial t},\tag{38}$$

where φ is the *electric scalar potential*. In electrostatics the second term is absent and φ is the familiar potential whose difference between two points is a voltage.

The gain is economy. The two source-free equations are now satisfied automatically and only the two with sources remain to be solved, and the six components of $\vec{E}$ and $\vec{B}$ are traded for four, φ and $\vec{A}$. The potentials are not unique: adding ∇f to $\vec{A}$ while subtracting $\partial f / \partial t$ from φ leaves both fields exactly as they were, for any f . That freedom is called *gauge invariance*, and it is used to impose whichever extra condition makes a given problem easiest. The potentials are also what enter the Lagrangian of a charged particle, and hence its quantum mechanics, so they are more than a calculational convenience.

12 The electric field in material media; dielectric polarisation, conductivity, mobility of charge carriers

12.1 Polarisation

Dielectric polarisation is the formation of electric dipoles, or the alignment of ones already present, under an applied electric field. The dipoles set up a field of their own inside the material which partly cancels the applied field, so the field inside a dielectric is always weaker than the field applied to it. Averaged per unit volume the dipole moments give the polarisation vector $\vec{P}$, whose linear form and role in Maxwell's equations are set out in [Section 11.3](#).

The response is only approximately linear. In general $\vec{P}$ is expanded in powers of the field, written here for one direction,

$$P = \varepsilon_0 \left(\chi^{(1)} E + \chi^{(2)} E^2 + \chi^{(3)} E^3 + \dots \right), \quad (39)$$

where $\chi^{(1)} \equiv \chi$ — the linear electric susceptibility of [Eq. 33](#). The higher coefficients are so small that the extra terms matter only in intense laser fields, where they are the whole of *nonlinear optics*: $\chi^{(2)}$ lets a crystal double the frequency of the light passing through it (second-harmonic generation), and $\chi^{(3)}$ makes the refractive index depend on the intensity. Strictly each $\chi^{(k)}$ is a tensor, since $\vec{P}$ need not point along $\vec{E}$ in an anisotropic crystal, which is the origin of birefringence.

Keeping the linear term alone and substituting it into the definition of the displacement field ([Eq. 34](#)) gives

$$\vec{D} = \varepsilon_0 \vec{E} + \vec{P} = \varepsilon_0 (1 + \chi) \vec{E} \equiv \varepsilon_0 \varepsilon_r \vec{E}, \quad (40)$$

where $\varepsilon_r \equiv 1 + \chi$ — relative permittivity (dielectric constant) of the material, the quantity actually tabulated. It is 1 in vacuum, a few for ordinary plastics and glasses, and about 80 for water at low frequencies. The three mechanisms below add up to it, and each stops contributing above its own characteristic frequency, which is why ε_r falls in steps as the frequency rises.

Electronic polarisation

Electronic polarisation is the displacement of the negatively charged electron cloud of an atom with respect to its positive nucleus. It occurs in all materials. Because only the light electrons have to move it is the fastest mechanism, and it still responds at optical and ultraviolet frequencies. The induced dipole moment is fixed by the electronic polarisability α_e ,

$$p_e = \alpha_e E.$$

Ionic polarisation

Ionic polarisation occurs only in materials with ionic bonds. It is the displacement of the ions of the crystal lattice, those of one sign one way and those of the other sign the other way. Whole atoms bound in a lattice must move, so it is far slower than the electronic mechanism and dies out above the infrared.

Orientational polarisation

Orientational polarisation, also called dipolar polarisation, occurs only in polar dielectrics, that is, in materials whose molecules carry a permanent dipole moment of their own. The field exerts an ordering torque on them, while thermal motion destroys the ordering, so this contribution alone depends on temperature and falls as the temperature rises. Whole molecules must reorient, so it is the slowest mechanism of all and stops following the field somewhere in the microwave range — which is exactly what a microwave oven exploits in water.

12.2 Conductivity and mobility

Conduction is the flow of electric charge through a material under an applied field. The carriers do not accelerate freely, because they are scattered by lattice vibrations and by defects, so in a steady field they settle at an average *drift velocity* v_d proportional to the field. The constant of proportionality is the *mobility*,

$$\mu \equiv \frac{v_d}{E}, \quad (41)$$

which measures how easily a carrier is pushed through a particular material. The current density is then $J = qnv_d = \sigma E$, the local form of Ohm's law, with the *conductivity*

$$\sigma = qn\mu, \quad (42)$$

where q is the charge of one carrier and n their concentration. Only two things can therefore change a conductivity: how many carriers there are, and how freely each moves. The most common kinds of conduction are electronic, in which the carriers are electrons in the conduction band, hole conduction, in which the carriers are the vacancies left in an otherwise full valence band and behave as positive charges, and ionic, in which whole ions move.

Temperature acts on both factors, and which one wins distinguishes the two standard cases. In a metal n is fixed, while more lattice vibrations mean more scattering and a smaller μ , so the conductivity falls as the material is heated. In a semiconductor n rises steeply with temperature, because carriers must first be excited across the band gap, and that easily outweighs the loss of mobility, so the conductivity rises instead.

13 The magnetic field in material media; classification of magnetic media, microscopic explanations of the exchange interaction between the magnetic moments of the constituents (atoms, ions, molecules)

On the microscopic scale every medium has magnetic properties, on account of the spin of its constituent particles and of the motion of the electrons about the nuclei. On the macroscopic scale magnetic behaviour is far less common, because these moments are not ordered over long distances and largely cancel one another. Long-range order can be imposed by an external magnetic field, just as polarisation is imposed by an electric one (Section 12).

The bookkeeping is also the same. The average magnetic dipole moment per unit volume is the magnetisation $\vec{M}$, and it is hidden inside the redefined field $\vec{H} \equiv \vec{B}/\mu_0 - \vec{M}$ (Eq. 35), which is sourced by the free currents we control and by nothing else (Section 11.3). Taking the response to be linear in $\vec{H}$,

$$\vec{M} = \chi_m \vec{H} \implies \vec{B} = \mu_0 (1 + \chi_m) \vec{H} \equiv \mu_0 \mu_r \vec{H}, \quad (43)$$

where χ_m — magnetic susceptibility, and $\mu_r \equiv 1 + \chi_m$ — relative permeability, the exact counterpart of $\epsilon_r = 1 + \chi$ (Eq. 40). One difference is worth noting: χ is positive for every ordinary dielectric, whereas χ_m may have either sign, and its sign is what classifies the material. For most media $|\chi_m| \sim 10^{-5}$, so magnetically they are almost vacuum, while in a ferromagnet μ_r reaches 10^3 or more and the linear relation fails altogether. Interactions with a magnetic field are also more subtle than with an electric one: by the Bohr–van Leeuwen theorem a purely classical system in equilibrium has zero mean magnetisation, so every magnetic effect below is a quantum one. Three kinds of medium are usually distinguished.

13.1 Paramagnets

The most common group. They have positive susceptibility, so the induced magnetisation points along the applied field. It comes from unpaired electrons, whose magnetic moments turn to line up with an external field. Common paramagnets are aluminium, platinum and oxygen. Two standard models:

- Curie paramagnetism — a set of independent atomic moments that do not interact with one another, as in the paramagnetic salts of transition-metal or rare-earth ions. Balancing their alignment against the randomising effect of heat gives Curie’s law for the susceptibility,

$$\chi_m = \frac{C}{T},$$

where T is the absolute temperature and C (the Curie constant) is fixed by the size and number of the atomic moments. The susceptibility falls as the temperature rises because thermal agitation fights the alignment.

- Pauli paramagnetism — in a metal the conduction electrons (as in an ordinary metal such as sodium) are treated as a *Fermi gas*: a gas of electrons that barely interact, but which (because no two electrons may share the same quantum state) fill the available energy levels from the bottom up to a top level called the Fermi level. Only the few electrons near that top level are free to reorient in a field, so the resulting paramagnetism is weak and almost independent of temperature.

13.2 Diamagnets

Diamagnetism is the appearance of a magnetisation *opposite* to the applied field, i.e. a negative susceptibility. Every material has this response, but it is very weak, so we call a material a diamagnet only when it is the dominant effect. That happens when the atoms carry no permanent magnetic moment—all electron spins paired—so there is no paramagnetic contribution to mask it. Typical diamagnets are bismuth (the strongest ordinary one), copper, gold and water.

A special case is that of superconductors, which expel the magnetic field from their interior completely through the Meissner effect. The remaining diamagnets are described by:

- Langevin diamagnetism — describes insulators (such as water and the inert gases), where the electrons are bound in atoms rather than free to conduct. An applied field makes each bound electron orbit precess (Larmor precession, [Section 6.3](#)). A precessing orbit is effectively a small current loop, and the magnetic moment it produces points against the applied field, which is the origin of the diamagnetic response.
- Landau diamagnetism — treats the same Fermi gas of conduction electrons in a metal as Pauli paramagnetism, but now accounts for the field bending the electrons' orbital motion rather than flipping their spin. This orbital effect is diamagnetic.

13.3 Ferromagnets

A third category appears once we include the *exchange interaction*—a quantum effect that makes neighbouring moments prefer to point the same way as each other, not just as the external field. Regions of aligned moments (magnetic domains) form, giving a strong susceptibility and *remanent* magnetisation: magnetisation that survives after the external field is switched off, which is what makes a permanent magnet. The classic ferromagnets are iron, cobalt and nickel.

Such materials are analysed with statistical-physics lattice models that differ in how freely each moment may point: the Ising model allows only up or down, the XY model allows any direction in a plane, and the Heisenberg model allows any direction in space (the most general).

Closely related are antiferromagnets and ferrimagnets, in which neighbouring moments point in *opposite* directions. In an antiferromagnet the opposing moments are equal and cancel, so there is almost no net magnetisation (manganese oxide and

chromium are examples). In a ferrimagnet they are opposite but unequal, so the cancellation is only partial and a net magnetisation remains—macroscopically it behaves like a weaker ferromagnet (magnetite is the classic example).

14 Electromagnetic waves as examples of solutions of Maxwell's equations; polarisation, reflection and refraction of waves, the Doppler effect

14.1 Electromagnetic waves as examples of solutions of Maxwell's equations

An electromagnetic wave is a solution of the source-free Maxwell equations in vacuum (Eq. 32). To show it, take the curl of Faraday's law, $\nabla \times \vec{E} = -\partial \vec{B} / \partial t$:

$$\nabla \times (\nabla \times \vec{E}) = \nabla \times \left(-\frac{\partial \vec{B}}{\partial t} \right).$$

We make use of the following identity:

$$\nabla \times (\nabla \times \vec{F}) = \nabla(\nabla \cdot \vec{F}) - \nabla^2 \vec{F}.$$

Then, we have:

$$\begin{aligned} \text{LHS} &= \nabla \times (\nabla \times \vec{E}) = \nabla(\nabla \cdot \vec{E}) - \nabla^2 \vec{E} = \nabla(0) - \nabla^2 \vec{E} = -\Delta \vec{E} \\ \text{RHS} &= \nabla \times \left(-\frac{\partial \vec{B}}{\partial t} \right) = -\frac{\partial}{\partial t} (\nabla \times \vec{B}) = -\frac{\partial}{\partial t} \left(\mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t} \right) = -\mu_0 \epsilon_0 \frac{\partial^2 \vec{E}}{\partial t^2}, \end{aligned}$$

where, for the LHS, we have used the identity and the first Maxwell equation, and for the RHS, we have used the fourth Maxwell equation and

$$\Delta \vec{F} = \nabla^2 \vec{F} = \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) \vec{F}$$

is the Laplacian operator.

Equating both sides we obtain:

$$\left(\Delta - \mu_0 \epsilon_0 \frac{\partial^2}{\partial t^2} \right) \vec{E} = 0 \tag{44}$$

We proceed analogously with the fourth Maxwell equation and find the equation for $\vec{B}$.

$$\left(\Delta - \mu_0 \varepsilon_0 \frac{\partial^2}{\partial t^2}\right) \vec{B} = 0$$

Both of these equations are standard forms of the wave equation, generally written as:

$$\left(\Delta - \frac{1}{v^2} \frac{\partial^2}{\partial t^2}\right) \vec{F} = 0,$$

where v is the propagation speed. From this it follows that the speed of propagation of an electromagnetic wave in vacuum is

$$v = c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}}.$$

Performing the same derivation inside a homogeneous, non-magnetic medium carrying no free charge, the sources do not all cancel and we are left with

$$\left(\Delta - \mu_0 \varepsilon_0 \frac{\partial^2}{\partial t^2}\right) \vec{E} = \underbrace{\mu_0 \frac{\partial \vec{J}_f}{\partial t}}_{\text{important in conductors}} + \underbrace{\mu_0 \frac{\partial^2 \vec{P}}{\partial t^2}}_{\text{important in insulators}},$$

check that this is general and correct physically

both terms being zero in vacuum. In a transparent insulator the first term is absent, and substituting the linear response $\vec{P} = \varepsilon_0 \chi \vec{E}$ (Eq. 33) collects the second one back into the left-hand side, replacing ε_0 by $\varepsilon_0 \varepsilon_r$. The wave equation is recovered with a smaller speed $v = c/n$, where

$$n \equiv \frac{c}{v} = \sqrt{\varepsilon_r \mu_r} \simeq \sqrt{\varepsilon_r} \quad (45)$$

is the *refractive index*, the quantity that appears in Snell's law (Eq. 46), and the approximation holds because $\mu_r \simeq 1$ for anything transparent. Since each polarisation mechanism stops following the field above its own frequency (Section 12), ε_r and hence n depend on frequency, which is *dispersion* and is why a prism separates colours.

Solution to the wave equation

The wave equation is solved by a plane wave of the form

$$\vec{E}(\vec{r}, t) = \vec{E}_0 e^{i(\vec{k} \cdot \vec{r} - \omega t + \phi)},$$

with the physical field understood as the real part, using $e^{i\alpha} = \cos \alpha + i \sin \alpha$. Here $\vec{k}$ points along the direction of travel, and for this transverse wave the constant

vector $\vec{E}_0$ is perpendicular to $\vec{k}$. The magnetic field $\vec{B}$ is perpendicular to both, with $|\vec{B}| = |\vec{E}|/c$. The direction in which $\vec{E}_0$ points is called the *polarisation* of the wave, discussed next.

Check with
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notes

14.2 Polarisation of a wave

Polarisation describes the pattern that the electric-field vector traces out, in the plane perpendicular to the direction of propagation, as the wave passes. Only transverse waves can be polarised, since the oscillation must have a choice of directions perpendicular to the travel direction (a longitudinal wave such as sound has none). Take propagation along z , so that the field lies in the x - y plane and the plane wave above reads

$$\vec{E}(z, t) = \text{Re} \left\{ \begin{bmatrix} E_x \\ E_y \end{bmatrix} e^{i(kz - \omega t)} \right\},$$

where E_x and E_y are complex. That pair of complex amplitudes is the *Jones vector*, and it fixes the polarisation completely through their relative size and relative phase. Three cases matter:

- **Linear** — the two components are in phase, so $\vec{E}$ stays along a fixed line. Along x the Jones vector is $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$, and at 45° to the axes it is $1/\sqrt{2} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$. Both entries are real, and a real Jones vector always means linear polarisation.
- **Circular** — the components have equal size but are a quarter of a period out of phase, so the tip of $\vec{E}$ traces a circle. The Jones vector is $1/\sqrt{2} \begin{bmatrix} 1 \\ \pm i \end{bmatrix}$, and the factor $\pm i$ is that quarter-period shift. Taking the upper sign at $z = 0$, the real parts are $E_x \propto \cos \omega t$ and $E_y \propto \text{Re}\{ie^{-i\omega t}\} = \sin \omega t$: the length of $\vec{E}$ never changes and the vector simply rotates. The sign of i fixes the sense of rotation, and the two are called left- and right-handed, though which is which depends on the convention being used.
- **Elliptical** — the general case, with unequal amplitudes or any other phase difference, tracing an ellipse. Linear and circular are its two limits.

The imaginary part is therefore not decoration: it is the only thing that distinguishes a rotating field from an oscillating one.

14.3 The laws of reflection and refraction of electromagnetic waves

A *ray* is a line drawn along the direction in which the wave (and its energy) travels, at right angles to the wavefronts. The laws below are stated in terms of rays.

The law of reflection

The angle of reflection equals the angle of incidence, and the incident ray, the reflected ray and the normal to the surface lie in one plane. Reflection changes only the direction of propagation, not the wavelength.

The law of refraction (Snell's law)

The incident and refracted rays and the normal lie in one plane, and the angles satisfy

$$n_1 \sin \theta_1 = n_2 \sin \theta_2, \quad (46)$$

where n_1 , n_2 are the refractive indices of the two media (Eq. 45), θ_1 the angle of incidence and θ_2 the angle of refraction, both measured from the normal.

To describe what happens to the polarisation on reflection, we fix the *plane of incidence*: the plane containing the incident ray and the normal. A wave is called *p*-polarised (parallel) when its electric field lies in this plane, and *s*-polarised (perpendicular, from the German *senkrecht*) when its field is perpendicular to it. The fractions of amplitude reflected and transmitted differ for the two, and are given by the Fresnel coefficients (forms omitted here).

The Brewster angle

The Brewster angle α_B is the angle of incidence at which the reflected and refracted rays are perpendicular to one another. At this angle the reflected *p*-component vanishes, so the reflected light is completely *s*-polarised. It is given by

$$\tan \alpha_B = \frac{n_2}{n_1}, \quad (47)$$

with the light incident from the side of medium n_1 (taking $n_1 < n_2$). This is how polarising sunglasses work, since light reflected from a road or a water surface arrives close to this angle and is largely polarised in one direction.

14.4 The Doppler effect

The *Doppler effect* is the difference between the frequency a source emits and the frequency an observer records when the two move relative to one another. For waves that travel in a medium (such as sound) the speeds of source and observer are measured relative to that medium, and the two cases are not symmetric. For light in vacuum, which needs no medium, only the relative velocity matters.

Acoustic (mechanical) waves

With v the wave speed in the medium, v_o the observer's speed and v_s the source's speed (both relative to the medium),

$$f' = f \frac{v \pm v_o}{v \mp v_s}, \quad (48)$$

where the upper signs apply when that party moves *towards* the other (raising the pitch) and the lower signs when it moves away. Observer and source enter differ-

ently: the observer changes the wave speed it meets, while the source changes the wavelength it emits.

Electromagnetic waves

Here the effect is relativistic and depends only on the relative speed through $\beta = v/c$ (Eq. 3). Writing ϕ for the angle between the relative velocity and the line of sight,

$$f' = f \frac{\sqrt{1 - \beta^2}}{1 \pm \beta \cos \phi}, \quad (49)$$

with the upper sign when the two are separating and the lower when they are approaching, exactly as in Eq. 48. Moving straight along the line of sight ($\phi = 0$) gives the longitudinal case,

$$f' = f \sqrt{\frac{1 \mp \beta}{1 \pm \beta}},$$

a redshift when separating and a blueshift when approaching. At $\phi = \frac{\pi}{2}$ the sign stops mattering and what remains is the *transverse* Doppler effect, $f' = f\sqrt{1 - \beta^2} = f/\gamma$, a pure redshift caused by time dilation that has no counterpart for mechanical waves.

15 Mechanical waves, electromagnetic waves, matter waves; similarities and differences

- **Electromagnetic waves** are oscillating electric and magnetic fields that satisfy the wave equation derived earlier (Eq. 44). They are transverse, travel at c in vacuum and need no medium. In the quantum picture the wave is a stream of photons, each with energy $E = \hbar\omega$ and momentum $p = \hbar k$, where ω is the angular frequency and $k = 2\pi/\lambda$ the wave number.
- **Mechanical waves** are disturbances passing through an elastic medium, as described in Section 9. They require a medium and may be longitudinal or transverse, with a speed set by the properties of the medium.
- **Matter waves** express the fact that every particle also behaves as a wave, with the (de Broglie) wavelength

$$\lambda = \frac{h}{p} = \frac{h}{mv} \quad (\text{non-relativistically}), \quad (50)$$

so the wave nature is easiest to see for light particles such as electrons, whose diffraction and interference are used in electron microscopy and diffraction imaging. A localised particle corresponds not to a single wavelength but to a *wave packet*—a superposition of many waves. The same relations $E = \hbar\omega$ and $p = \hbar k$ apply.

Common features and differences

All three show interference and diffraction, obey the laws of reflection and refraction, and can form standing waves by superposition. They differ in what they need and what they are: mechanical waves need a medium, whereas electromagnetic and matter waves do not. Electromagnetic and matter waves are quantum in nature—for a matter wave the squared amplitude gives the probability of finding the particle (*postulate II*), while for light it corresponds to the number of photons.

16 Phase transitions; kinds of phase transition, the order parameter of the transition, conditions for the coexistence of phases (examples of phenomena)

A *phase* is a region that is uniform in its physical and chemical properties—ice, liquid water and steam are three phases of the same substance. A *phase transition* is an abrupt change in these properties when a control variable such as temperature or pressure is varied. Transitions may involve the state of matter, the crystal structure, magnetic ordering or superconductivity.

To describe them we use a thermodynamic potential—for changes at fixed temperature and pressure this is the *Gibbs free energy*, the quantity a system minimises at equilibrium (it is not the enthalpy). It is written as a function of an *order parameter*: a quantity that is zero in the more symmetric, higher-entropy (disordered) phase and becomes nonzero in the ordered phase. Two examples are the magnetisation of a ferromagnet (zero above the Curie temperature, nonzero below) and the density difference between coexisting liquid and gas.

The *Ehrenfest classification* sorts transitions by how the chemical potential μ (the free energy per particle) behaves. A transition is of the n -th kind when the n -th derivative $\partial^n \mu / \partial T^n$ is the first that is discontinuous.

- First order — a first derivative jumps, producing latent heat and a sudden change of density. Melting and boiling are examples.
- Second order — the first derivatives are continuous but a second derivative jumps, with no latent heat. A ferromagnet losing its magnetisation at the Curie temperature is the standard example.

- Higher orders are not used in practice, and the scheme itself breaks down whenever a derivative *diverges* instead of jumping, as the heat capacity does at the superfluid transition of helium. Modern usage therefore keeps only two classes: first-order transitions, which have latent heat, and *continuous* transitions, which do not.

The conditions for phases to coexist are given by the *Gibbs phase rule*,

$$f = s - r + 2, \quad (51)$$

where s is the number of chemical components, r the number of coexisting phases and f the number of intensive variables (such as temperature and pressure) that may be changed while keeping those phases in equilibrium. It requires the system to be in thermal, mechanical and chemical equilibrium. For a single substance ($s = 1$):

- $r = 1$: one phase, $f = 2$ — an area on the pressure–temperature diagram.
- $r = 2$: two phases coexisting, $f = 1$ — a coexistence curve.
- $r = 3$: three phases, $f = 0$ — the triple point.

17 Experimental foundations of quantum mechanics; examples of phenomena that can be explained in quantum mechanics, but not in classical theory

Quantum mechanics was forced on physics by a handful of experiments that classical theory gets not merely inaccurately but qualitatively wrong. The postulates themselves are set out in [Section 5](#), and the origin of discrete energies in [Section 5.6](#). What follows is the evidence for them.

Black-body radiation

A hot body radiates with a spectrum fixed by its temperature alone. Classically every mode of the field should carry the same average energy, and since the number of modes grows without limit towards short wavelengths, the predicted power diverges—the *ultraviolet catastrophe*, which is a failure of the classical prediction and not something anyone observed. The measured spectrum instead peaks and falls away. In 1900 Planck reproduced it by assuming that a mode of frequency ν exchanges energy only in whole quanta $E = h\nu$, which makes the short-wavelength modes too expensive to excite ([Eq. 59](#)).

The photoelectric effect

Light falling on a metal ejects electrons. Classically their energy should grow with the intensity of the light. What is measured is that the kinetic energy of each electron depends only on the *frequency*, while the intensity fixes only the *number* of electrons, and that below a threshold frequency nothing is emitted however bright the light. This follows if the light arrives as quanta of energy $h\nu$, one photon freeing one electron against the work function W that binds it to the metal,

$$E_{\text{kin}} = h\nu - W.$$

The Compton effect

X-rays scattered from weakly bound electrons come off with a longer wavelength, the shift growing with the scattering angle θ ,

$$\Delta\lambda = \frac{h}{m_e c} (1 - \cos \theta).$$

A classical wave would scatter without changing its wavelength. The result follows at once if the photon is treated as a particle carrying both energy and momentum, so that the collision is an ordinary elastic one and the photon gives up a share of each.

Atomic spectra and the stability of the atom

Atoms absorb and emit only at sharp, characteristic frequencies, and the Franck–Hertz experiment shows the same discreteness directly, electrons passing through a vapour losing energy only in fixed amounts. Classical physics allows a continuous range of orbits and so no sharp lines at all, and worse, it cannot keep the atom together: an orbiting electron is an accelerating charge, so it would radiate and spiral into the nucleus almost at once. Both problems disappear once the allowed energies are discrete, which is not an extra assumption but a consequence of the boundary conditions on a bound wave function ([Section 5.6](#)).

Interference of single particles

Coherent light through two slits gives interference fringes, as any wave would. The quantum result is that electrons fired one at a time build up the same pattern, each arriving as a single localised impact at a point that cannot be predicted. The wave function fixes only the probability, and the fringes come from amplitudes rather than probabilities being added, which is worked through in [Section 5](#). Merely making it possible to tell which slit was used destroys the pattern, and nothing classical behaves in that way.

Other purely quantum phenomena

Several more appear elsewhere in this note and are worth naming in the same breath: tunnelling through a barrier the particle has not the energy to cross (Section 21), the two-valued spin of the electron and its splitting of a beam in the Stern–Gerlach experiment (Section 6), the exclusion of identical fermions from a common state, which is what gives matter its volume and the periodic table its structure (Eq. 61), and entanglement, in which a pair of particles has a state but neither member has one of its own (postulate VII).

18 The uncertainty principle and the limits of measurement accuracy

The Heisenberg uncertainty principle states that certain pairs of observables cannot both have a definite value in the same state: the more precisely one is fixed, the less precisely the other can be. This is a property of the quantum state itself, not a shortcoming of our instruments.

Whether two observables $\hat{A}$ and $\hat{B}$ can be definite together is decided by their *commutator*

$$[\hat{A}, \hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A},$$

which measures how much the order of the two operations matters. If it vanishes the observables commute and can be measured simultaneously to any accuracy. If it does not, the spreads of the two quantities obey the Robertson relation

$$\boxed{\Delta A \Delta B \geq \frac{1}{2} |\langle [\hat{A}, \hat{B}] \rangle|} \quad (52)$$

Here ΔA is the ordinary standard deviation of the measured values, built from the expectation values of Section 5,

$$\Delta A = \sqrt{\langle \hat{A}^2 \rangle - \langle \hat{A} \rangle^2}, \quad \langle \hat{A} \rangle = \langle \Psi | \hat{A} | \Psi \rangle,$$

that is, the typical distance of a single result from the mean of many. It is zero exactly when the state is an eigenstate of $\hat{A}$, where every run returns the same eigenvalue.

The most important case is position and momentum. Since $[\hat{x}, \hat{p}_x] = i\hbar$, the relation gives

$$\boxed{\Delta x \Delta p_x \geq \frac{\hbar}{2}}, \quad (53)$$

saturated (turned into an equality) by a Gaussian wave packet, which is therefore the best-localised state possible in position and momentum at once. The energy–time relation $\Delta E \Delta t \geq \hbar/2$ used in [Section 2](#) has the same form but a different reading, since time is a parameter and not an observable with an operator: there Δt is how long the state persists, not the spread of a measured quantity.

The trade-off is easiest to see for a wave. A pure sine wave has a single wavelength, so by the de Broglie relation ($p = \hbar k$ with $k = 2\pi/\lambda$, [Eq. 50](#)) its momentum is perfectly defined—but it extends over all space, so its position is completely undefined. A sharply localised spike has a definite position, but building it requires adding together waves of every wavelength, so its momentum is completely undefined. Any real state is a compromise between these extremes.

This should not be confused with ‘Heisenberg’s microscope’, the argument that pinning down a particle’s position needs light of ever shorter wavelength, whose photons then carry ever larger momentum and so disturb the particle more. That picture is misleading, because it still assumes the particle has a definite position waiting to be measured, which is exactly what the principle denies.

19 Approximate methods of quantum mechanics; perturbation theory for eigenvalues, Fermi’s golden rule, the Born approximation for scattering amplitudes

Quantum mechanics describes the behaviour of systems at the atomic and subatomic level. However, it rests to a large extent on solving differential equations which we are able to solve analytically only in relatively simple cases. Sometimes even determining the Hamiltonian itself is difficult, and therefore for more complex systems approximate methods are used, which make it possible to obtain useful results.

19.1 Stationary perturbation theory

Given a known Hamiltonian H_0 , for example that of the hydrogen atom, we may wish to introduce a slight modification into the system, for example a weak external electric field. Solving it analytically again would be just as highly non-trivial as the hydrogen atom itself. We may instead treat our correction as a perturbation in the form of an additional term $\lambda H'$. The method itself is the quantum counterpart of an expansion in a Taylor series with respect to the parameter λ and of calculating

successive corrections. The smaller the perturbation, the fewer corrections need to be calculated, and often one or two suffice. They are given by the following formulae:

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$$E_n^{(1)} = \langle \psi_n^{(0)} | H' | \psi_n^{(0)} \rangle \quad (54)$$

$$\psi_n^{(1)} = \sum_{m \neq n} \frac{\langle \psi_m^{(0)} | H' | \psi_n^{(0)} \rangle}{E_n^{(0)} - E_m^{(0)}} \psi_m^{(0)} \quad (55)$$

$$E_n^{(2)} = \sum_{m \neq n} \frac{|\langle \psi_m^{(0)} | H' | \psi_n^{(0)} \rangle|^2}{E_n^{(0)} - E_m^{(0)}} \quad (56)$$

19.2 Fermi's golden rule

The method of stationary perturbation theory described above assumes that the perturbing potential is invariant in time. There also exists a dynamical method, but it is far more complicated. In the special case, however, in which the perturbing potential varies periodically in time, one may apply Fermi's golden rule, an important result of that calculation. This solution gives the probability per unit time of transitions between two quantum states $\psi_n^{(0)}$ and $\psi_m^{(0)}$, for a perturbation $H'e^{-i\omega t}$:

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$$\Gamma_{n \rightarrow m} \simeq \frac{2\pi}{\hbar} |\langle \psi_m^{(0)} | H' | \psi_n^{(0)} \rangle|^2 \delta(E_m^{(0)} - E_n^{(0)} - \hbar\omega) \quad (57)$$

The delta function enforces energy conservation: the transition happens only if the quantum $\hbar\omega$ supplied by the perturbation matches the gap between the two levels.

19.3 The Born approximation

A further application of perturbation theory is the Born approximation for scattering. A particle is scattered off a fixed, spherically symmetric potential $V(r)$. The incident particle is described by a plane wave, and far from the potential the state is this incident wave plus an outgoing spherical wave whose angular amplitude $f(\theta, \phi)$ is the *scattering amplitude*. Its squared modulus $|f|^2$ is the differential cross-section, that is, how much is scattered into each direction.

Writing the Schrödinger equation in an equivalent integral form (the Lippmann–Schwinger equation) and keeping only its first iteration gives the (first) Born approximation. For a spherically symmetric potential,

$$f(\theta) = -\frac{2m}{\hbar^2 K} \int_0^\infty r' V(r') \sin(Kr') dr', \quad (58)$$

where $K = |\vec{k} - \vec{k}'| = 2k \sin(\theta/2)$ is the momentum transfer (the change in wave vector on scattering) and k is the wave number. It works best for weak potentials or fast particles, where the scattering is a small perturbation. (This is unrelated to the Born–Oppenheimer approximation of molecular physics, which separates the slow nuclear motion from the fast electronic motion.)

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20 The Schrödinger equation as the basis for understanding the structure of the atom and of the chemical molecule; the periodic table of the elements, chemical bonds

20.1 The atom from the Schrödinger equation

A state evolves in time by the Schrödinger equation (Eq. 18), and the states of definite energy are the solutions of its time-independent form (Eq. 17). For one electron bound to a nucleus of charge Ze the potential is the Coulomb attraction

$$V(r) = -\frac{Ze^2}{4\pi\epsilon_0 r},$$

and because it depends on the distance alone the solutions split into a radial and an angular factor, $\psi_{nlm}(r, \theta, \varphi)$. Physically acceptable solutions—finite, single-valued, normalisable—exist only for particular integer labels, the *quantum numbers* (Section 6):

- **Principal**, $n = 1, 2, 3, \dots$ — fixes the energy and roughly the size, $E_n = -13.6 Z^2 \text{ eV} / n^2$ for a one-electron atom, so 13.6 eV is the energy needed to remove the electron from hydrogen in its ground state.
- **Orbital**, $l = 0, \dots, n - 1$ — fixes the magnitude of the orbital angular momentum, $|\vec{L}| = \hbar\sqrt{l(l+1)}$, and hence the shape. Written as the letters s, p, d, f.
- **Magnetic**, $m = -l, \dots, +l$ — fixes its projection on a chosen axis, $L_z = m\hbar$, that is, the orientation.
- **Spin**, $m_s = \pm 1/2$ — the projection of the electron's own spin, doubling every count above.

One such solution is an *orbital*: a spatial probability distribution for one electron. Orbitals sharing the same n form a *shell*, and those sharing both n and l a *subshell*. Since the exclusion principle (Eq. 61) forbids two electrons the same four labels, one orbital holds two electrons of opposite spin, a subshell holds $2(2l+1)$ —two in s, six in p, ten in d, fourteen in f—and a shell holds $2n^2$.

No exact solution exists once there is more than one electron, because each also repels the others, but the labels survive and the ground state is built by filling the levels according to three rules:

- **Aufbau principle** — fill from the lowest energy upwards, in order of increasing $n + l$, and for equal $n + l$ in order of increasing n . This is why 4s fills before 3d.
- **Pauli exclusion principle** — no two electrons in one atom share all four quantum numbers.

- **Hund's rule** — within a subshell, electrons occupy separate orbitals with parallel spins before any orbital takes a second electron.

The result is the *electron configuration*, written for carbon as $1s^2 2s^2 2p^2$. Only the outermost, *valence* electrons are available for bonding, and they are what the chemistry of the element is made of.

Corrections to this picture

The energies above are the *gross structure*. Smaller effects split each level further, and it is worth keeping their sizes in order. The *fine structure* comes from relativistic corrections and from the electron's magnetic moment sitting in the field produced by its own orbital motion (spin-orbit coupling), and it is smaller than the gross structure by a factor of order $\alpha^2 \simeq 5 \times 10^{-5}$, where $\alpha \simeq 1/137$ is the fine-structure constant. The *hyperfine structure* comes from the far weaker magnetic moment of the nucleus and is smaller again by roughly the electron-to-proton mass ratio—in hydrogen it splits the ground state by about 6×10^{-6} eV, the transition responsible for the 21 cm radio line used to map galaxies. An external field splits levels in its turn, into the Zeeman effect for a magnetic field and the Stark effect for an electric one, both of which are worked out later in the note.

20.2 The periodic table of the elements

The *periodic table* arranges all the chemical elements in order of increasing *atomic number* Z —the number of protons in the nucleus, which fixes the identity of the element and equals the number of electrons in the neutral atom. (The *mass number* A , the total number of protons and neutrons, is a separate quantity and is not what orders the table.) The modern *periodic law* states that the properties of the elements vary periodically with atomic number. Mendeleev originally ordered the elements by atomic mass, which gives almost the same sequence, but the correct ordering is by atomic number.

The layout of the table is the filling order written out, so the shape of the table is a direct consequence of the rules above:

- a **group** (vertical column) collects elements with a similar valence-shell configuration, and hence similar chemical behaviour. The noble gases close a subshell and react with almost nothing.
- a **period** (horizontal row) corresponds to the principal quantum number n of the outermost occupied shell, and its length—2, 8, 8, 18—is the capacity of the subshells being filled along it.

20.3 Molecules and the different types of chemical bond

The same equation applied to two nuclei rather than one explains why atoms stick together. As two atoms approach, each pair of atomic orbitals combines into two *molecular orbitals*: a *bonding* one, in which the two wave functions add and pile electron density up between the nuclei, where it screens their repulsion and lowers the energy, and an *antibonding* one, in which they cancel between the nuclei and the

some later section mentioned molecular bonding which is not here. fix somewhere

energy is raised. A bond exists when more electrons sit in bonding orbitals than in antibonding ones. Everything below is a special case of valence electrons rearranging in this way to lower the total energy.

- **Ionic** — one atom transfers electrons to another, and the resulting positive and negative ions attract electrostatically. It is strong and non-directional, and it forms crystalline solids (e.g. NaCl).
- **Covalent** — neighbouring atoms share pairs of electrons in a common bonding orbital so that each attains a filled valence shell. It is strong and directional, since the orbitals point in particular directions, which is what fixes the shapes of molecules (e.g. H₂, diamond).
- **Metallic** — in a metal the atoms are packed so closely that the outermost shell of each overlaps the shells of many neighbours at once. The valence electrons are then not tied to any particular pair of atoms and move freely through the whole solid, leaving positive ion cores in a shared sea of delocalised electrons. Because no bond points anywhere in particular, the ions can slide past one another without the bonding being broken, which is why metals are malleable, and because the electrons are free to move they conduct heat and electricity (e.g. copper).
- **Hydrogen bond** — a much weaker attraction, in which a hydrogen atom already covalently bound to a strongly electronegative atom (O, N, F) is drawn towards a second such atom nearby, the hydrogen being left with little electron density of its own. It holds liquid water together and binds the two strands of DNA.
- **Van der Waals forces** — the weakest of all, acting between whole neutral molecules with no electron sharing and no charge transfer, and falling off quickly with distance (as r^{-6}). They come in three kinds, all dipole–dipole: between two permanent dipoles, between a permanent dipole and the dipole it induces in a neighbour, and between the fluctuating instantaneous dipoles of two atoms whose electron clouds happen to be lopsided at the same moment (the London dispersion force). Only the last is available to a noble gas, and it is what allows even helium to be liquefied.

21 Quantum tunnelling; examples

21.1 The barrier problem

Quantum tunnelling is the passage of a particle through a potential barrier higher than the particle's own energy—something forbidden in classical mechanics, where the particle simply lacks the energy to get over. The barrier must be finite in both

height and width. Against an infinitely high wall the wave function is forced to zero at the boundary and nothing passes, which is why the infinite square well of [Section 5.6](#) confines a particle perfectly and the finite well does not.

Consider a particle of energy E meeting a barrier of height $V_0 > E$ and width b . Solving the Schrödinger equation in the three regions gives:

- **before the barrier** — an incident plus a reflected wave, $\psi(x) = Ae^{ikx} + Be^{-ikx}$.
- **inside the barrier** — since $E < V_0$ the wave does not oscillate but decays exponentially, $\psi(x) = Ce^{-\kappa x} + De^{\kappa x}$, with $\kappa = \sqrt{2m(V_0 - E)}/\hbar$.
- **beyond the barrier** — a transmitted wave travelling outwards, $\psi(x) = Fe^{ikx}$, with a smaller amplitude, $|F| < |A|$.

Because the exponential tail inside the barrier is still nonzero at the far edge, the transmitted amplitude does not vanish: there is a finite probability of the particle emerging. This is measured by the transmission coefficient

$$T = \frac{|F|^2}{|A|^2} \propto e^{-2\kappa b},$$

the proportionality holding for any barrier thick enough to be worth calling one. Everything about tunnelling follows from that exponential. The probability falls off extremely fast with the width b of the barrier and with $\sqrt{m(V_0 - E)}$, so tunnelling is a serious effect only for light particles and thin, low barriers, and small changes in any of these quantities change the rate by orders of magnitude.

21.2 Examples

- **Alpha decay.** An α particle inside a heavy nucleus is held by the strong interaction but faces the Coulomb repulsion of the daughter nucleus as a barrier several MeV higher than its own energy. It escapes by tunnelling. Because T depends exponentially on the energy, a spread of only a few MeV in α energies produces half-lives ranging from microseconds to billions of years, which no other mechanism would explain.
- **Fusion in stars.** Two nuclei must come within about 10^{-15} m against their mutual Coulomb repulsion. At the temperature of the Sun's core the typical thermal energy is a thousand times smaller than that barrier, so on classical grounds the Sun should not shine. Fusion proceeds because the rare fast protons tunnel through, and this is also why controlled fusion may be possible at a lower temperature than a classical estimate demands.
- **Scanning tunnelling microscope.** A sharp conducting tip is held a fraction of a nanometre above a surface and a small voltage applied. Electrons tunnel across the gap, and because the current depends exponentially on the width of that gap, moving the tip by the diameter of a single atom changes it by orders of magnitude. Scanning the tip at constant current maps the surface atom by atom.

- **Semiconductor devices.** In a tunnel diode the junction is made so narrow that carriers tunnel straight across it, which gives the device a range of voltages over which the current *falls* as the voltage rises. In flash memory, writing and erasing consist of driving electrons through a thin insulating layer onto an isolated gate, where they stay until driven back.
- **The ammonia molecule.** The nitrogen atom of NH_3 sits to one side of the plane of the three hydrogens and tunnels back and forth through it. The two arrangements have the same energy, and the tunnelling splits the ground level into two, the transition between them being the microwave line on which the first maser was built.

22 Physical foundations of optical spectroscopy; electronic, vibrational and rotational excitations of molecules and the corresponding ranges of the electromagnetic spectrum

Optical spectroscopy is an analytical technique which studies the interaction of electromagnetic radiation with matter. By analysing the absorption, emission or scattering of light by atoms and molecules, one obtains information about their electronic, vibrational and rotational structure.

A molecule can store energy in three ways, and they differ greatly in the amount of energy involved. The larger the energy of a transition, the higher the frequency (and the shorter the wavelength) of the photon absorbed or emitted, since $E = hf = hc/\lambda$. In decreasing order of energy these are electronic, vibrational and rotational excitations, each about one to two orders of magnitude below the one above it. In molecules all three are present at once, so a single electronic transition carries a whole comb of vibrational and rotational sub-levels on top of it, and this is what turns what would be one atomic line into a broad structured band.

A transition is only seen if the light can couple to it, which for absorption and emission means that the motion must change the electric dipole moment of the molecule. This is why a symmetric molecule such as N_2 has no infrared spectrum at all, while CO_2 and H_2O do, and it is the whole basis of the greenhouse effect.

22.1 Electronic excitations

Electronic excitations are jumps of an electron between the bound energy levels of an atom or molecule (Section 20). When a photon of the right energy is absorbed, an electron moves from a lower to a higher level, which shows as a line or band in the absorption spectrum. These are the largest steps, a few electronvolts, which puts them in the visible and near-ultraviolet, roughly 200–800 nm. They are also what makes things coloured, since the colour we see is the part of the visible spectrum that was *not* absorbed.

22.2 Vibrational excitations

Vibrational excitations are oscillations of the atoms within a molecule about their equilibrium positions, for example the stretching or bending of a bond. Any potential minimum is harmonic for small displacements, so the levels are those of a harmonic oscillator, $E_v = (v + 1/2) \hbar \omega$ with $v = 0, 1, 2, \dots$ (Section 5.6), and are therefore evenly spaced. The spacing is of order 0.1 eV, one to two orders of magnitude below an electronic transition, which places them in the infrared, roughly 2.5–25 μm . Chemists quote the same thing as a wavenumber, $1/\lambda = 400\text{--}4000 \text{ cm}^{-1}$, and each functional group has its own characteristic band, which is what makes infrared spectroscopy an identification tool.

22.3 Rotational excitations

Rotational excitations are changes in the rotation of the whole molecule about its centre of mass, so they change its angular momentum, which is quantised as in Section 6. The levels are $E_J = BJ(J + 1)$ with $J = 0, 1, 2, \dots$, so unlike the vibrational ladder they are not evenly spaced but spread out as J grows, and the constant B is inversely proportional to the moment of inertia — heavier molecules have more closely spaced levels. These are the smallest steps of the three, of order a few meV down to much less, giving wavelengths from a fraction of a millimetre to a few centimetres: the far infrared and the microwave region. Rotational spectra are the sharpest of the three and measure bond lengths directly through B , though a spectrum with all three kinds of structure at once can be dense and hard to disentangle.

23 Phenomena accompanying the interaction of electromagnetic radiation with atoms and molecules; stimulated and spontaneous emission, absorption

An atom or molecule with discrete energy levels can exchange energy with light in three basic ways: it can absorb a photon, emit one spontaneously, or be triggered to

emit one by a passing photon. All three are *resonant*, meaning they occur only when the photon energy hf matches the gap between two levels, and each therefore probes the level scheme directly. Light also interacts off resonance, by *scattering*: elastically, with the photon merely redirected and its energy unchanged (Rayleigh scattering, which is stronger for short wavelengths and is why the sky is blue), or inelastically, with the molecule left in a different vibrational state so that the scattered photon has shifted in frequency (the Raman effect, which reaches vibrations that infrared absorption cannot). A further everyday case is fluorescence, in which a molecule absorbs a high-energy photon, loses a little energy without radiating, and emits a photon of lower energy.

23.1 Absorption

Absorption is the taking up of the energy of an electromagnetic wave by a substance, which reduces the intensity of the transmitted beam and lifts an atom or molecule to a higher level.

In cold, rarefied atomic gases the excited levels are well separated, so only photons of a few sharply defined energies can be absorbed. If white (continuous-spectrum) light passes through such a gas, dark lines appear in the transmitted spectrum at exactly those energies. This is how the composition of a stellar atmosphere is read off.

In molecules the level scheme is richer, because on top of the electronic levels there are the vibrational and rotational levels of [Section 22](#). These lie so close together that they merge into broad absorption *bands* rather than sharp lines.

23.2 Spontaneous emission

Spontaneous emission is the emission of a photon when an atom in an excited level drops, of its own accord, to a lower level. It happens without any external trigger and at a random moment, and it accounts for almost all everyday luminescence (heated gases, excited atoms, light-emitting diodes, and so on). The direction and phase of the photon are random, so the light is incoherent. With no re-excitation the number of atoms still excited decays exponentially,

$$N(t) = N(0) e^{-t/\tau},$$

where τ is the mean lifetime of the excited state.

23.3 Stimulated emission

Stimulated emission is the emission of a photon triggered by an incoming photon whose energy equals the level gap. The triggering photon is *not* absorbed, it only sets the process off. The emitted photon is a copy of it: same frequency, phase, polarisation and direction of travel. A beam built from such identical photons is *coherent*, and this multiplication of identical photons is the physical basis of the laser.

23.4 The Einstein coefficients

The three processes are put on one footing by writing down their rates per atom. Let $\rho(f)$ be the energy density of the radiation per unit frequency at the transition frequency, and let levels 1 and 2 be the lower and the upper one. Then absorption goes at $B_{12}\rho(f)$, stimulated emission at $B_{21}\rho(f)$, and spontaneous emission at A_{21} alone, with no ρ in it, because it needs no incident light. The three constants are the *Einstein coefficients*.

Requiring these rates to balance for atoms in equilibrium with black-body radiation (Eq. 59) fixes two relations between them,

$$g_1 B_{12} = g_2 B_{21}, \quad \frac{A_{21}}{B_{21}} = \frac{8\pi h f^3}{c^3},$$

where g_1 and g_2 count the states at each level. Two conclusions follow, and they are the reason the formalism is worth quoting. First, absorption and stimulated emission are equally likely per photon, so which one wins depends only on which level holds more atoms. In thermal equilibrium the lower one always does, so a beam is always attenuated. Amplification requires a *population inversion*, $N_2 > N_1$, which cannot occur in equilibrium and has to be created by pumping the medium, and which needs at least three levels to sustain. Second, the ratio A_{21}/B_{21} grows as f^3 , so spontaneous emission swamps stimulated emission at high frequency. That is why the maser came before the laser and why an X-ray laser is hard to build.

24 Atoms in strong electric and magnetic fields (the Stark effect, the Zeeman effect); experimental methods of studying spin

The atomic levels and quantum numbers used throughout this section, and the ordering of the splittings by size, are set out in Section 20. Both effects below are treated as perturbations of the atomic Hamiltonian (Section 19).

look online for simpler explanation

The Stark effect

The Stark effect is the splitting or shifting of atomic energy levels in an external electric field. We take the field along z , so the extra potential energy of the electron is

$$\vec{E} = E\hat{e}_z \rightarrow \hat{H}' = eEz = eEr \cos \theta,$$

a perturbation added to the atomic Hamiltonian (perturbation theory and the first-order shift are set out in [Eq. 54](#)).

For the hydrogen *ground state* $|100\rangle$ the first-order shift vanishes:

$$E_{100}^{(1)} = \langle 100 | \hat{H}' | 100 \rangle = eE \int \psi_{100}^* r \cos \theta \psi_{100} d^3r = 0,$$

because the ground state is spherically symmetric, so the positive and negative z contributions cancel. The ground state therefore shifts only at *second* order, proportionally to E^2 (the quadratic Stark effect).

Excited levels behave differently when they are degenerate. The $n = 2$ states $|200\rangle$ and $|210\rangle$ have the same energy and are mixed by the field, so here we must use *degenerate* perturbation theory: diagonalise $\hat{H}'$ within the $n = 2$ manifold. The only non-zero matrix elements are

$$\langle 200 | \hat{H}' | 210 \rangle = \langle 210 | \hat{H}' | 200 \rangle = -3eEa_0,$$

so that in the basis $\{|200\rangle, |210\rangle, |211\rangle, |21-1\rangle\}$

$$\hat{H}' = \begin{bmatrix} 0 & -3eEa_0 & 0 & 0 \\ -3eEa_0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Its eigenvalues are $\lambda_1 = \lambda_2 = 0$ and $\lambda_3 = -\lambda_4 = 3eEa_0$, and the two states that split off are

$$\frac{1}{\sqrt{2}} (|200\rangle \pm |210\rangle),$$

with energies

$$E = E_{200}^{(0)} \pm 3eEa_0.$$

This shift is linear in the field (the linear Stark effect), and it appears already at first order precisely because the level was degenerate.

The Zeeman effect

The Zeeman effect is the splitting of atomic energy levels in an external magnetic field. Many atomic levels are degenerate, meaning several distinct states share the same energy, distinguished only by quantities such as the orbital angular momentum or the spin. A magnetic field couples differently to states of different magnetic moment and so lifts this degeneracy, splitting one spectral line into several closely spaced ones.

Taking the field along z , the perturbation is $\hat{H}' = -\mu_z B$, where the magnetic moment of the electron is

$$\vec{\mu} = -\frac{\mu_B}{\hbar} (\vec{L} + g_S \vec{S}),$$

with μ_B the Bohr magneton and $g_S \simeq 2$ the electron spin g -factor.

If the spin plays no role ($S = 0$), we have the *normal* Zeeman effect. Since L_z has eigenvalue $\hbar m_l$, the $\hbar$ cancels and the shift is evenly spaced:

$$E_{nlm_l}^{(1)} = \langle nlm_l | \hat{H}' | nlm_l \rangle = \mu_B B m_l.$$

If the spin does contribute ($S \neq 0$) we have the *anomalous* Zeeman effect, with a more complicated pattern of lines. In a weak field the level splits according to the total angular momentum $\vec{J} = \vec{L} + \vec{S}$ with the Landé factor g_J :

$$E^{(1)} = \mu_B g_J B m_j, \quad g_J = 1 + \frac{j(j+1) + s(s+1) - l(l+1)}{2j(j+1)}.$$

Only in a *strong* field, where $\vec{L}$ and $\vec{S}$ effectively decouple (the Paschen–Back regime), does the shift simplify to the naive form $\mu_B B (m_l + 2m_s)$.

Experimental methods of studying spin

The best-known method is the Stern–Gerlach experiment. A beam of atoms (originally silver) is sent through a strongly *inhomogeneous* magnetic field. Any particle with non-zero spin has a magnetic moment whose projection on the field axis is quantised, and in an inhomogeneous field this moment feels a net force that deflects the atom by an amount set by that projection. The beam therefore splits into a discrete number of sub-beams. For silver, quantum mechanics predicts exactly two spots on the screen, corresponding to the two projections of a spin- $\frac{1}{2}$ electron, and this is what is observed (broadened only by the finite beam width and the spread of atomic speeds).

[Check online](#)

Three further methods are worth naming:

- **Magnetic resonance.** A static field splits the spin states by $\Delta E = g\mu_B B$, and a transverse radio-frequency field drives transitions between them when $hf = \Delta E$, which shows up as a sharp absorption peak. The condition is the same as saying that the driving frequency matches the Larmor precession of the spin (Section 6.3). Applied to electrons this is electron spin resonance, applied to nuclei it is nuclear magnetic resonance, and the latter is the basis of magnetic resonance imaging.
- **Zeeman spectroscopy.** Counting the lines a spectral line splits into, and measuring their spacing, gives g_J and hence j , l and s for the level. Historically it was the anomalous pattern, unexplained by orbital motion alone, that pointed to the existence of spin in the first place.

- **Precision measurement of g .** A single electron held in a Penning trap gives g_S to a dozen digits. Its small departure from 2 is the sharpest test of quantum electrodynamics there is.

25 Thermal radiation; the emissive and absorptive power of bodies, Kirchhoff's law, black-body radiation

Thermal radiation is the electromagnetic radiation emitted by every body at a temperature above absolute zero, arising from the thermal motion of its charged particles. It spans a wide range of wavelengths, from the infrared through the visible and into the ultraviolet as the temperature rises. The key idealisation is the *black body*: a perfect absorber and emitter, which absorbs all radiation falling on it regardless of wavelength, direction or polarisation, and emits with the greatest possible efficiency for its temperature. The nearest thing to one in the laboratory is a small hole in the wall of a heated cavity, since anything entering it is almost certain to be absorbed before it finds its way out again.

25.1 Emissive and absorptive power

Emissive power $E(f, T)$ measures how strongly a body emits: the energy radiated per unit area, per unit time, per unit frequency, at frequency f and temperature T .

Absorptive power $A(f, T)$ is the fraction of incident power that the body absorbs (the rest being reflected or transmitted), so it lies between 0 and 1.

25.2 Kirchhoff's law

Kirchhoff's law of thermal radiation states that at a fixed temperature the ratio $\varepsilon(f, T) = E(f, T)/A(f, T)$ is the same universal function for *all* bodies. Since a black body has $A = 1$, this universal function is simply the emissive power of the black body. In words: a good absorber at some frequency is necessarily an equally good emitter at that frequency, since otherwise a body placed in a cavity at its own temperature would spontaneously heat up or cool down, in breach of the second law (Section 8).

25.3 The black body and Planck's law

The calculation has two factors. First, count the standing-wave modes of the field in a cavity: their number per unit volume and per unit frequency grows as f^2 , since ever more ways of fitting a wave into the box appear as the wavelength shortens. Second, give each mode its average energy. Classical equipartition (Section 29) gives every mode the same $k_B T$ regardless of frequency, and the two factors together give

the Rayleigh–Jeans law, $\varepsilon \propto f^2 k_B T$. Since the mode count never stops growing, the predicted power diverges towards short wavelengths and the total radiated power is infinite. This is the *ultraviolet catastrophe*, and it is a failure of the classical prediction rather than an observed effect — the measured spectrum peaks and then falls away.

Planck kept the mode count and replaced the average energy. If a mode of frequency f can only take energy in whole quanta hf , then a mode with $hf \gg k_B T$ cannot be excited at all, because there is not enough thermal energy to buy even one quantum. Boltzmann-weighting the ladder of allowed energies gives an average of $hf/(e^{hf/k_B T} - 1)$ in place of $k_B T$, and hence

$$\varepsilon(f, T) = \frac{2\pi f^2}{c^2} \frac{hf}{e^{hf/k_B T} - 1}. \quad (59)$$

The two limits show what has happened. When $hf \ll k_B T$ the exponential may be expanded and the bracket returns $k_B T$, recovering Rayleigh–Jeans, so the classical result survives at low frequency. When $hf \gg k_B T$ the exponential dominates and the emission is cut off exponentially, which removes the divergence. This was the first appearance of the quantum of action anywhere in physics (Section 17).

Two limiting laws follow from Eq. 59: *Wien’s displacement law*, giving the wavelength of peak emission, $\lambda_{\max} = b/T$ with $b \simeq 2.9 \times 10^{-3}$ m K, so that hotter bodies peak at shorter wavelengths, and the *Stefan–Boltzmann law*, giving the total power radiated per unit area, $\Phi = \sigma T^4$ with $\sigma \simeq 5.7 \times 10^{-8}$ W m⁻² K⁻⁴.

26 Diffraction and interference of waves; coherence, diffraction of radiation on crystals

26.1 Interference and coherence

Interference is the superposition of waves. Where two waves meet in phase they reinforce (a bright fringe), where they meet out of phase they cancel (a dark fringe). This produces a stable pattern only if the waves are *coherent*, meaning the phase difference between them at a given point stays constant in time. An ordinary lamp emits independent wave trains from independent atoms (Section 23), so its phase is scrambled within a very short time and no pattern survives, whereas a laser is coherent by construction.

The classic demonstration is Young’s double-slit experiment. Light from a single source passes through one slit and then through two further slits a distance d apart, which act as two point sources of coherent spherical waves:

$$\begin{aligned}\psi_1(r_1, t) &= A(r_1) \cos(kr_1 - \omega t) \\ \psi_2(r_2, t) &= A(r_2) \cos(kr_2 - \omega t).\end{aligned}$$

Adding them gives a wave whose amplitude is modulated by a factor $\cos(k(r_1 - r_2)/2)$. This factor is largest (a *bright fringe*) when the two path lengths differ by a whole number of wavelengths,

$$r_1 - r_2 = m\lambda,$$

and vanishes (a *dark fringe*) when they differ by an odd number of half-wavelengths,

$$r_1 - r_2 = \frac{2m+1}{2}\lambda,$$

with m an integer. The same experiment performed with single particles is one of the foundations of quantum mechanics (Section 17).

26.2 Diffraction

There is no sharp distinction between diffraction and interference. By convention, the pattern from a *finite number* of discrete coherent sources is called interference, and that from a *continuous* distribution of sources is called diffraction. Physically, diffraction is the bending of waves around obstacles or through apertures: by Huygens' principle every point of a wavefront acts as a source of new spherical wavelets, and their superposition beyond the obstacle (taking amplitudes and phases into account) gives the diffracted pattern. The effect is only noticeable when the obstacle or aperture is comparable in size with the wavelength, which is why sound bends around a doorway and light does not.

26.3 Diffraction of radiation on crystals

A *crystal* is a solid whose atoms sit in a pattern that repeats periodically in all three directions — long-range order, described by a lattice of equivalent points together with the group of atoms placed at each of them. Nothing in that definition refers to the bonding: the bond holding the atoms in place may be ionic (NaCl), covalent (diamond), metallic (copper) or molecular (ice), and it is the arrangement alone that makes the solid a crystal (Section 20). A solid without such long-range order, glass being the standard case, is called *amorphous*, and it gives broad haloes instead of sharp diffraction spots.

molecular bonding was not defined earlier

The regular spacing acts as a three-dimensional diffraction grating, and it works only for radiation whose wavelength is comparable with the atomic spacing of about 10^{-10} m — X-rays, or equally the matter waves of neutrons and electrons of the right energy (Eq. 50). When such radiation falls on a crystal it scatters off the atoms, and because these sit on parallel planes a fixed distance d apart, the waves scattered

from successive planes interfere. They reinforce only when the extra path travelled between neighbouring planes is a whole number of wavelengths, which is Bragg's law:

$$n\lambda = 2d \sin \theta, \quad (60)$$

where n is the diffraction order, λ the wavelength, d the interplanar spacing and θ the glancing angle (measured between the beam and the crystal plane, not the normal). Strong diffracted beams therefore appear only at particular angles, and measuring those angles gives the spacings and hence the structure. This is how the arrangement of atoms in a solid, and of the atoms in DNA and in proteins, was determined.

Although the reinforced beam emerges at an angle equal to the angle of incidence, so that the effect is often loosely called 'reflection', it differs from ordinary reflection: it uses the whole three-dimensional stack of atomic planes rather than just the surface, occurs only at the special Bragg angles rather than at every angle, and yields a much weaker beam than a good mirror.

27 Indistinguishability of elementary particles; the connection between spin and statistics, Bose–Einstein condensation

Identical particles (electrons, quarks, photons and so on) cannot be told apart, and this is not merely a practical limitation of our instruments. In quantum field theory each kind of particle is an *excitation of a single underlying field*: every electron is a ripple in the one electron field, much as two middle-C notes sounded on identical strings are the same excitation of the same kind of system rather than two notes that merely happen to match. There is therefore no hidden label that could distinguish one electron from another, and this has a sharp consequence for the wave function. Write the wave function of N such particles as $\psi(1, 2, \dots, N, t)$, where each number stands for the position and spin coordinates of one particle. Exchanging two particles cannot change any measurable quantity, so it can at most multiply the wave function by a phase:

$$\psi(1, 2, \dots, N, t) = e^{i\varphi} \psi(2, 1, \dots, N, t). \quad (61)$$

verify that the analogy with guitar is good

Swapping the same pair a second time returns the original configuration, which forces $e^{2i\varphi} = \pm 1$. Hence the wave function of identical particles must be either completely *symmetric* (+) or completely *antisymmetric* (−) under exchange.

The spin–statistics postulate

Which sign applies is fixed by the spin. Particles of half-integer spin $(\frac{1}{2}, \frac{3}{2}, \dots)$, the *fermions*, have an antisymmetric wave function. Particles of integer spin $(0, 1, 2, \dots)$, the *bosons*, have a symmetric one. An immediate corollary for fermions is the *Pauli exclusion principle*: an antisymmetric wave function vanishes if two particles share the same state, so no two fermions can occupy the same quantum state. This is what stacks electrons into shells and gives the periodic table its shape (Section 20).

Bose–Einstein condensation

Bosons are not bound by the exclusion principle, so any number of them may share one state. For a gas of non-interacting bosons the mean occupation of a state of energy ε_k is the Bose–Einstein distribution (Eq. 68),

$$n_k = \frac{1}{e^{(\varepsilon_k - \mu)/k_B T} - 1},$$

where μ is the chemical potential, k_B the Boltzmann constant and T the temperature. (Here the label k is the wave vector of the state, not to be confused with k_B .) Because of the -1 in the denominator, the occupation of the lowest state can grow without limit. Below a critical temperature a large fraction of the particles pile into that single lowest-energy state and behave collectively, almost as one object. This is *Bose–Einstein condensation*. It is not condensation in ordinary space (the atoms are not gathered in one place) but in momentum space, since so many particles come to share the same momentum. The critical temperature is

$$T_c \simeq 3.31 \frac{\hbar^2 n^{2/3}}{m k_B},$$

with n the number density of atoms and m the mass of one atom. It is extremely low for a dilute gas, of order 10^{-7} K for the alkali vapours in which the condensate was first made in 1995, because the condensation sets in only once the thermal de Broglie wavelength (Eq. 50) grows to the spacing between the atoms and their waves overlap. Superfluidity and superconductivity are the same phenomenon in disguise, the latter for pairs of electrons bound into composite bosons.

28 Statistical ensembles; classical and quantum statistics, thermodynamic potentials and their relationship to the statistical sums of respective ensembles

A statistical ensemble is a large imaginary collection of copies of a system, each in some microscopic state (microstate) consistent with the same fixed macroscopic conditions. Averaging over the ensemble reproduces the macroscopic (thermodynamic) properties. The three standard ensembles differ in which macroscopic quantities are held fixed, and each is tied to a particular thermodynamic potential through its partition function.

A *partition function* is a weighted count of the microstates available to the system: weighted, because when the energy is not fixed a microstate of higher energy is exponentially less likely. It does two jobs. It is the denominator that normalises the probability of a microstate, and it is the generating function of the whole thermodynamics: every macroscopic quantity is obtained from it by differentiation, for instance the mean energy as $-\partial \log Q / \partial \beta$ and the pressure as $k_B T \partial \log Q / \partial V$. It is the bridge across which a list of microscopic states becomes a thermodynamic potential.

28.1 The microcanonical ensemble

Here the number of particles N , the volume V and the energy E are all fixed. Every accessible microstate is equally likely, and the partition function Ω is just the total number of them. Boltzmann linked this count to the entropy:

$$S = k_B \log \Omega. \quad (62)$$

28.2 The canonical ensemble

Fixing the energy exactly is awkward, so instead we let the system exchange heat with a reservoir at fixed temperature T (with N and V still fixed). The microstates are then weighted by the Boltzmann factor, and the partition function is a sum over all microstates r :

$$Q_N(V, T) = \sum_r \exp(-E_r / k_B T). \quad (63)$$

It yields directly the Helmholtz free energy,

$$A = -k_B T \log Q_N. \quad (64)$$

28.3 The grand canonical ensemble

Taking one further step, we also let the particle number vary, fixing instead the chemical potential μ (with V and T). Summing over all microstates *and* all particle numbers gives

$$Q = \sum_{r,s} \exp(-\alpha N_r - \beta E_s), \quad (65)$$

with $\beta = 1/k_B T$ and $\mu = -\alpha/\beta$. This connects to the grand thermodynamic potential by

$$\Xi = -k_B T \log Q. \quad (66)$$

28.4 The thermodynamic potentials

The internal energy U is the natural quantity when the system is isolated, and the first law (Eq. 26) written for a reversible change reads $dU = T dS - p dV + \mu dN$. Its natural variables are therefore S , V and N . That is inconvenient, because nobody in a laboratory holds entropy fixed — what is easy to hold fixed is temperature, or pressure. Each potential below is U with one such awkward variable traded for the quantity conjugate to it, by subtracting the product of the pair.

- **Enthalpy** $H = U + pV$, natural in S , p , N . At constant pressure and with no work other than expansion, $dH = dU + p dV = \delta Q$: the heat taken up in a constant-pressure process *is* the change in enthalpy. Since almost all chemistry happens in an open vessel at atmospheric pressure, this is why reaction heats are tabulated as enthalpies.
- **Helmholtz free energy** $A = U - TS$, natural in T , V , N — exactly the variables of the canonical ensemble, which is why Eq. 64 takes the form it does.
- **Gibbs free energy** $G = U - TS + pV$, natural in T , p , N . These are the two variables an experimenter actually controls, which is why G governs phase equilibria and chemical reactions (Section 16).
- **Chemical potential** $\mu = (\partial G / \partial N)_{T,p}$, the energy cost of adding one more particle to the system.

What makes the free energies the quantities to minimise is the second law applied to the system *and* its surroundings. Let a system be held at temperature T by a reservoir. The reservoir loses the heat the system gains, so its entropy changes by $-\delta Q/T$, and the total entropy cannot decrease:

$$dS_{\text{total}} = dS - \frac{\delta Q}{T} \geq 0 \implies \delta Q \leq T dS.$$

At fixed T and V no work is done, so $\delta Q = dU$ and the condition becomes $dU - T dS = dA \leq 0$. The Helmholtz free energy of a system at fixed temperature

what does it mean to add a particle to the system? why shooting an atom into a container would cost any energy?

and volume can only fall, and equilibrium is where it stops falling, that is, at its minimum. Repeating the argument at fixed T and p , where the system also does work $p dV$ against the surroundings, gives $dG \leq 0$ in the same way. So each potential is minimised under precisely the conditions in which it is natural, and the familiar rule that entropy increases is the special case for an isolated system.

The same inequality is what makes the name ‘free’ energy literal rather than decorative. If the system delivers work δW at constant temperature, then $\delta W = \delta Q - dU \leq T dS - dU = -dA$. The work that can be extracted is at most the drop in A , and not the drop in U : of the internal energy, the portion TS is tied up in keeping the system’s entropy where it is and must be surrendered to the surroundings as heat. What is left over, $U - TS$, is the part free to become work, and the same argument at constant p makes $-\Delta G$ the ceiling on work of any kind other than pushing the atmosphere aside. The chemical potential plays the analogous role for matter: just as temperature equalises when heat can flow and pressure equalises when volume can be exchanged, μ equalises when particles can be exchanged, and a difference in μ is what drives diffusion, evaporation and every phase change.

28.5 Quantum statistics

The same machinery works in quantum mechanics, but for identical particles ([Section 27](#)) the allowed occupations of each energy state are further restricted. The quantity to compare is the mean number of particles $\langle n_\varepsilon \rangle$ occupying a state of energy ε , and the three statistics differ only in what they permit.

Classically, particles are treated as though they could be labelled, and any number may sit anywhere: the occupation falls off as a plain exponential in the energy. For fermions no state may hold more than one particle, so the occupation can never exceed 1. At absolute zero the states are filled from the bottom up to an energy called the *Fermi level*, and μ is that level: the distribution is a step, and raising the temperature only rounds off the step over a width of about $k_B T$. This is why only the small fraction of electrons lying within $k_B T$ of the Fermi level can respond to anything at all, and hence why the electronic contribution to a metal’s heat capacity and to its paramagnetism ([Section 13](#)) is so much smaller than a classical estimate. For bosons there is no such restriction, the occupation of a low-lying state may grow without bound as ε approaches μ , and that runaway is Bose–Einstein condensation. In all three cases, whenever the states are so sparsely occupied that $\langle n_\varepsilon \rangle \ll 1$, the ± 1 in the denominators below stops mattering and the quantum distributions collapse onto the classical one — which is the dilute, hot limit in which classical statistical physics is safe.

1. **Maxwell–Boltzmann** (classical): treats particles as distinguishable. It is a good approximation at high temperature and low density, but fails for a dense gas of electrons or photons.
2. **Fermi–Dirac**: for fermions, which obey the Pauli exclusion principle (at most one particle per state):

$$\langle n_\varepsilon \rangle = \frac{1}{e^{\beta(\varepsilon - \mu)} + 1}. \quad (67)$$

3. **Bose–Einstein**: for bosons, which are indistinguishable but not exclusion-bound:

$$\langle n_\varepsilon \rangle = \frac{1}{e^{\beta(\varepsilon - \mu)} - 1}. \quad (68)$$

29 Statistical explanation of the macroscopic properties of systems of very many particles: extensive and intensive parameters of the state of a system, the equation of state, specific heats (of gases and solids), magnetic susceptibilities, etc.

Classical thermodynamics is not itself a statistical theory. It is phenomenological: it defines quantities such as temperature, entropy and pressure by how they behave, states a few laws relating them (Section 8), and never mentions atoms. Everything it says is therefore true whatever matter turns out to be made of, which is its strength, but it cannot predict the value of anything — an equation of state or a heat capacity has to be measured and fed in. Statistical physics supplies exactly that missing half. It assumes matter is made of very many particles obeying mechanical laws, and shows that the thermodynamic quantities are averages over their microstates, computed from a partition function (Section 28). Two things follow. The laws of thermodynamics are explained rather than postulated — entropy becomes a count of microstates (Eq. 62), and the second law becomes the statement that a system drifts to the macrostate realised by overwhelmingly the most microstates. And the numbers thermodynamics leaves open can be calculated from the microscopic model, which is what the rest of this section illustrates.

is that right?

29.1 Extensive and intensive parameters

State parameters come in two kinds:

- **intensive** — independent of the amount of material, e.g. pressure, temperature, density and any ‘per-mole’ quantity.
- **extensive** — proportional to the amount of material, e.g. volume, mass, particle number, total internal energy and total entropy.

The distinction is microscopic in origin. Cut a large system in two: the number of microstates of the whole is the product of the numbers for the halves, so the entropy, being a logarithm of that count, adds. Anything built by adding contributions particle by particle is extensive, while the intensive quantities are the ratios and derivatives of extensive ones and describe the local condition of the material rather than how much of it there is.

29.2 The equation of state

An equation of state relates the state parameters of a system, and may be written schematically as

$$p = p(\rho, T, s, u, \dots).$$

The two most common are the ideal-gas law and the van der Waals equation (the latter in a consistent n -mole form):

$$pV = nRT \quad (69)$$

$$\left(p + \frac{an^2}{V^2}\right)(V - nb) = nRT. \quad (70)$$

The correction b accounts for the finite volume of the molecules, and an^2/V^2 for their mutual attraction by the van der Waals forces of [Section 20](#). Statistically, both equations come out of one calculation: evaluate the partition function for a gas and differentiate it with respect to volume. Free point particles give [Eq. 69](#) exactly, and putting the intermolecular potential back in gives corrections of which [Eq. 70](#) is the first and crudest. That the attraction term is the one that permits liquefaction, and hence a liquid–gas transition ([Section 16](#)), is the point of including it.

29.3 Specific heats

Specific heat is the heat needed to raise the temperature of unit mass by one degree,

$$c = \frac{\Delta Q}{m \Delta T}.$$

The heat capacity measured while holding some quantity x fixed is $c_x = (\partial Q / \partial T)_x$. The two usual cases are constant volume and constant pressure. For one mole of a monatomic ideal gas,

$$c_V = \frac{3}{2}R, \quad c_P = \frac{5}{2}R,$$

so that $c_P - c_V = R$ (the extra energy at constant pressure goes into the work of expansion).

Statistically, a heat capacity counts the ways a molecule can store energy. Classical equipartition gives $\frac{1}{2}k_B T$ to every quadratic term in the energy, so three directions of translation give $\frac{3}{2}R$ per mole, and a diatomic gas, which can also rotate about two axes, gives $\frac{5}{2}R$. In a solid each atom is held in place and can both move and be displaced against a restoring force, which is six quadratic terms, so $c = 3R \simeq 25 \text{ J mol}^{-1} \text{ K}^{-1}$ — the Dulong–Petit law, and it is obeyed at room temperature by most simple solids.

Where this reasoning fails is itself informative. Measured heat capacities fall towards zero as the temperature does, and a diatomic gas behaves as though its vibrations, and eventually its rotations, were simply absent. The reason is quantisation: a degree of freedom whose level spacing exceeds $k_B T$ cannot be excited at all and is ‘frozen out’, contributing nothing. Equipartition is the high-temperature limit in which the spacing is negligible. The Einstein and Debye models put this right for a solid, the latter giving the observed $c \propto T^3$ at low temperature, while in a metal a further term linear in T survives from the few conduction electrons near the Fermi level (Section 28).

29.4 Magnetic susceptibility

For a magnetic system the work done in changing the magnetisation is $dW = \mu_0 H dM$, where M is the total magnetic moment. The magnetisation is the moment per unit volume, $m = M/V$. The isothermal and adiabatic magnetic susceptibilities measure how the moment responds to the applied field:

$$\chi_T = \frac{1}{\mu_0} \left(\frac{\partial M}{\partial H} \right)_T, \quad \chi_S = \frac{1}{\mu_0} \left(\frac{\partial M}{\partial H} \right)_S.$$

Microscopically the susceptibility is a competition. Each atomic moment has a lower energy pointing along the field than against it, so Boltzmann weighting favours alignment, while thermal agitation randomises the directions. Averaging the moment over that weighting gives a magnetisation proportional to B/T and hence Curie’s law $\chi_m = C/T$ (Section 13). It is a good illustration of the general pattern: a macroscopic response function is a thermal average of a microscopic quantity, and its temperature dependence records which of the two effects is winning.

30 The Dirac and Klein–Gordon equations; the form of the equations and examples of objects governed by each of them

The Schrödinger equation (Eq. 18) cannot be the whole story, because it was built from the non-relativistic $E = p^2/2m$ and therefore treats time and space quite differently: one derivative in time, two in space. Relativity requires them to enter on the same footing (Section 1). The two equations below are the two ways of repairing this, and they differ in the spin of the particles they end up describing. Both start from the relativistic energy–momentum relation

$$E^2 = p^2 c^2 + m^2 c^4. \quad (71)$$

KG eqn is mentioned in QM1. Look it up

The Klein–Gordon equation

Substituting the usual operators $E \rightarrow i\hbar \partial/\partial t$ and $\vec{p} \rightarrow -i\hbar \nabla$ (Section 5) directly into Eq. 71 gives

$$\left(\square + \frac{m^2 c^2}{\hbar^2} \right) \psi(\vec{r}, t) = 0, \quad (72)$$

where $\square = \frac{1}{c^2} \partial_t^2 - \nabla^2$ is the d'Alembertian, the four-dimensional analogue of the Laplacian: it is the same combination of derivatives, with the time derivative included on the same footing as the three spatial ones, so it treats spacetime as a whole rather than space alone.

As to its *form*: it is second order in time and second order in space, and it is nothing other than the ordinary wave equation (Eq. 44) with a mass term added. Setting $m = 0$ returns the equation for light exactly, so a massless Klein–Gordon field simply travels at c , and the mass term is what slows it down and gives it a rest energy. The wave function is a single complex number at each point — one component, no internal structure — and that is the formal statement that it describes a particle of *spin 0*: it has nothing that could point anywhere.

Its solutions are plane waves $\psi \propto e^{i(\vec{k} \cdot \vec{r} - \omega t)}$ with $\varepsilon = \pm c \sqrt{p^2 + m_0^2 c^2}$, and the two historical difficulties are visible in that formula and in the order of the equation. The negative root gives states of negative energy, into which any particle could fall without limit, and being second order in time the equation does not yield a probability density that stays positive. Both objections dissolve in quantum field theory, where ψ is not a probability amplitude for one particle but a field whose excitations are particles, and the negative-energy solutions become antiparticles. Objects governed by it: spin-0 particles, such as the pion and the other spin-0 mesons, and the Higgs boson.

The Dirac equation

Dirac looked for an equation *first* order in time, so as to keep a sensible probability density, and therefore, by relativity, first order in space as well. That means taking a square root of Eq. 71, and a linear expression can only square to $p^2 c^2 + m^2 c^4$ if its coefficients anticommute, so that the unwanted cross terms cancel. Ordinary numbers cannot do this. Matrices can. For a free particle the result is

$$(i\hbar \gamma^\mu \partial_\mu - mc) \Psi(x^\nu) = 0. \quad (73)$$

Each object in it has a plain reading:

- ∂_μ is the *four-gradient*, the four-vector (Section 1) built from the derivatives with respect to the four coordinates $x^\nu = (ct, x, y, z)$, that is, $\partial_\mu = (\partial/\partial(ct), \partial/\partial x, \partial/\partial y, \partial/\partial z)$. It is the relativistic replacement for ∇ (Eq. 14), with time included as a fourth direction.
- An index appearing twice, as μ does here, is *summed over* its four values by convention, so $\gamma^\mu \partial_\mu$ is shorthand for a sum of four terms.

- The γ^μ are the four *gamma matrices*, 4×4 constant matrices chosen so that $\{\gamma^\mu, \gamma^\nu\} = 2g^{\mu\nu}I$, where $g^{\mu\nu}$ is the metric that fixes the relative sign of the time and space parts. That anticommutation condition is the whole content of ‘taking the square root’ above.
- Ψ is therefore not one number but a column of four, called a *spinor*: two components for the two spin orientations of the particle and two more for its antiparticle.

A common representation, written in 2×2 blocks, is

$$\gamma^0 = \begin{bmatrix} I & 0 \\ 0 & -I \end{bmatrix}, \quad \gamma^i = \begin{bmatrix} 0 & \sigma_i \\ -\sigma_i & 0 \end{bmatrix},$$

with σ_i the *Pauli matrices*, the three 2×2 matrices that represent the components of spin through $\hat{S}_i = \frac{\hbar}{2}\sigma_i$. Two of them have already appeared as $\hat{S}_z$ and $\hat{S}_x$ in [Section 5](#), so the spin structure of the Dirac equation is built out of an object already familiar.

Its form, then, is: first order in the derivatives, matrix-valued, and acting on a four-component object rather than on a single function. Objects governed by it: all spin- $\frac{1}{2}$ particles, that is, every particle of matter — electrons, muons, quarks — free or coupled to the electromagnetic field. Its successes were immediate. Spin was not put in by hand but emerged from the requirement of relativistic invariance alone, together with very nearly the right magnetic moment ($g_S \simeq 2$, [Section 24](#)). It reproduces the fine structure of the hydrogen spectrum. And the extra components it forced on the wave function predicted the positron, found four years later.

31 The structure of atomic nuclei and nuclear reactions; nuclear binding energy, fission and fusion reactions, alpha and beta decay, radioactive decay series, the limits of the world of nuclides

The atomic nucleus is made of protons and neutrons, collectively *nucleons*. The number of protons (Z) fixes the chemical element, and the number of neutrons (N) distinguishes its isotopes. The nucleons are held together by the strong nuclear force, far stronger than the electromagnetic force but of very short range (about 1–2 femtometres, [Section 2](#)).

A *nuclide* is one particular nuclear species, specified by both Z and N (equivalently by Z and the mass number $A = Z + N$). It is a finer notion than ‘element’, which fixes Z alone. Nuclides sharing Z are *isotopes* of one element, and about 250 of the roughly 3000 known nuclides are stable.

31.1 Binding energy and mass

The mass of a nucleus is always *less* than the sum of the masses of the nucleons composing it. The difference is the *mass defect*,

$$\Delta m = Zm_p + Nm_n - M(Z, N),$$

and the *binding energy* is that missing mass expressed as energy, $B = \Delta m c^2$ (Section 1). It is the energy needed to pull the nucleus apart into free nucleons, and it is missing from the mass precisely because it would have to be supplied to do so. This is mass–energy equivalence in its most direct experimental form, and it is why a nuclear reaction releases so much more energy than a chemical one: the binding involved is millions of times larger per particle.

What matters for stability is not B but B/A , the binding energy *per nucleon*. Plotted against A it gives the binding-energy curve, and almost all of nuclear energetics is contained in its shape. It rises steeply through the light nuclei, reaches a broad maximum of about 8.8 MeV per nucleon near $A \simeq 56$ –62 (iron and nickel), and then falls slowly, to about 7.6 MeV for uranium. A reaction releases energy whenever it moves nucleons to a higher point on this curve, and there are only two ways to do so. Light nuclei climb by *fusing* together, heavy nuclei climb by *splitting* apart, and both therefore run towards iron from their respective sides. Fusing hydrogen releases about 7 MeV per nucleon, whereas fissioning uranium recovers only the roughly 0.9 MeV per nucleon by which uranium sits below the peak — some 200 MeV per fission, but far less per nucleon than fusion.

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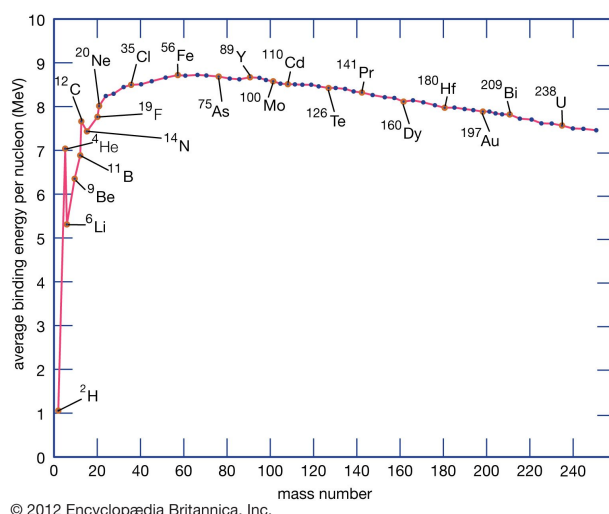


Figure 2: Binding energy per nucleon as a function of mass number A . The curve rises to a maximum near iron and nickel, and falls off slowly for heavier nuclei.

The curve itself is reproduced by the *semi-empirical mass formula* of the liquid-drop model, a sum of a volume term proportional to A , a surface term that penalises small nuclei, a Coulomb term that penalises large Z , an asymmetry term favouring

$N \simeq Z$, and a pairing term. The last of these is what puts the local jumps into an otherwise smooth curve: nuclei with even Z and even N are more tightly bound than their odd neighbours, which is why stable even- Z elements have many isotopes and odd- Z ones typically have one or two.

31.2 Structure and reactions

Two models capture the structure. The *liquid-drop model* treats the nucleus like a drop of incompressible fluid, with the strong force and the electrostatic repulsion playing the roles of surface tension and internal pressure. It reproduces the gross trend of the binding energy and explains why very heavy nuclei can split. The *shell model*, by analogy with atomic electron shells (Section 20), has nucleons filling discrete energy shells. It explains the deviations from the liquid-drop trend and the *magic numbers* 2, 8, 20, 28, 50, 82, 126, at which nuclei are especially tightly bound.

Nuclear reactions transform nuclei, often under bombardment by neutrons or protons:

- **fission** — a heavy nucleus splits into two (or more) lighter ones, releasing neutrons and a large amount of energy. The released neutrons can trigger further fissions, which is the chain reaction.
- **fusion** — two light nuclei merge into a heavier one, releasing energy, but only if they can be brought within range of the strong force against their electrostatic repulsion (Section 32).
- **alpha decay** — emission of an alpha particle (a helium-4 nucleus), which is itself a tunnelling process (Section 21).
- **beta decay** — emission of an electron (β^- , a neutron turning into a proton) or a positron (β^+ , the reverse), together with an (anti)neutrino, via the weak interaction.
- **electron capture** — the nucleus absorbs one of its own innermost atomic electrons, which converts a proton into a neutron and emits a neutrino. It competes with β^+ decay, has the same effect on Z and N , and is the only route open when there is too little energy available to create a positron.
- **gamma decay** — emission of a high-energy photon by an excited nucleus returning to its ground state, usually following one of the above.

A *radioactive decay series* is a chain of successive decays leading from one unstable nuclide to the next until a stable one is reached. There are four such series in principle, but only three occur naturally today (the thorium, uranium and actinium series). The fourth, the neptunium series, has effectively died out, because its longest-lived member (^{237}Np , half-life about 2.1 million years) is short-lived compared with the age of the Earth.

31.3 The limits of the world of nuclides

The limits are set by how many protons or neutrons a nucleus can hold before it becomes unbound (the proton and neutron ‘drip lines’), and, for the heaviest nuclei, by fission.

- **The proton drip line:** because the Coulomb barrier traps protons and lengthens the lifetime of proton-unbound nuclei, the proton-rich edge is comparatively easy to reach. It has been crossed as far as bismuth ($Z = 83$), though it is firmly established only up to about aluminium ($Z \simeq 13$).
- **The neutron drip line:** neutrons feel no Coulomb barrier, so the neutron-rich edge is far harder to pin down. It has been experimentally confirmed only up to neon ($Z = 10$), and mapping it further is a major current challenge.
- **Superheavy nuclei:** here the limit is spontaneous fission. Their stability rests on shell effects, and the existence of a predicted ‘island of stability’ of long-lived superheavy nuclides is still an open question.

32 Nucleosynthesis and the energy sources of stars

The elements are produced in four natural ways: primordial nucleosynthesis just after the Big Bang, fusion inside stars, neutron capture in and around dying stars, and spallation by cosmic rays.

The Big Bang: $Z \leq 3$

The first protons formed once the Universe had cooled enough for quarks to bind into hadrons (roughly 10^{-6} s to 1 s after the Big Bang). *Primordial nucleosynthesis* then followed (about 10 s to 10^3 s), producing the light nuclei: deuterium (^2H), tritium (^3H), helium (^3He and ^4He), lithium (^7Li) and beryllium (^7Be). It stopped there because no stable nuclide has $A = 5$ or $A = 8$, so there is no way to build upwards by adding one nucleon at a time, and the Universe thinned and cooled too fast for anything slower. The ^7Be seen today is not primordial, since its half-life is only about 53 days: it long ago turned into ^7Li by electron capture ([Section 31](#)).

Stars: Z up to 26

Stars formed much later and drove the next round. They fuse hydrogen into helium, then helium into carbon (three ^4He nuclei at once, the triple-alpha process, which is how the $A = 8$ gap is bridged), and in the more massive stars carbon, neon, oxygen and silicon burn in turn, each stage requiring a higher temperature than the last and building nuclei up to iron. There it stops. It is often said that fusion ‘stops

at iron' because ^{56}Fe is the most stable nucleus, but this needs care: ^{56}Fe has the smallest *mass per nucleon*, whereas the largest *binding energy per nucleon* belongs to nickel, ^{62}Ni (the two differ because the proton and neutron masses are not equal). In practice stellar fusion favours ^{56}Ni , which is less stable and decays to ^{56}Fe . Either way, past the peak of the binding-energy curve ([Section 31](#)) fusion would consume energy rather than release it, so it can no longer support the star.

Neutron capture: Z above 26

Everything heavier is built not by fusion but by adding neutrons, which feel no Coulomb repulsion and so need no barrier to be overcome, followed by β^- decay of the neutron-rich product. Two regimes are distinguished by how the capture rate compares with the decay rate. The *s-process* (slow) proceeds in the interiors of evolved low-mass stars, one neutron at a time with leisure to decay in between, and it builds the nuclides lying along the valley of stability up to bismuth. The *r-process* (rapid) dumps many neutrons on a nucleus before any decay can occur, requires an extreme neutron flux, and is responsible for the most neutron-rich heavy nuclides, gold, platinum, thorium and uranium among them. Supernovae were long assumed to be its site, and the merger of two neutron stars observed in 2017 confirmed at least one such site directly.

Cosmic rays: lithium, beryllium and boron

These three light elements are an exception on both counts: stars destroy them rather than make them, and the Big Bang made only a trace of lithium. They come instead from *spallation*, in which a fast cosmic-ray proton strikes a heavier nucleus in the interstellar medium and chips fragments off it.

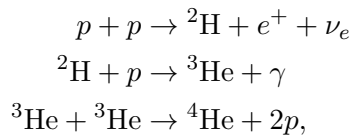
32.1 Thermonuclear reaction

A *thermonuclear reaction* fuses two light nuclei into a heavier one. Being exothermic, it releases energy (as kinetic energy of the products and gamma rays) that is then shared with the surroundings as heat. The energy released can be found from the mass defect ([Section 31](#)).

For hydrogen nuclei (protons) to fuse they must come within the range of the strong force (~ 1 fm), but they first have to overcome their mutual electrostatic repulsion, a barrier of order 1 MeV. Classically this needs a temperature of about 10^{10} K, far hotter than any stellar core. Fusion nonetheless proceeds at the much lower core temperatures because of *quantum tunnelling* ([Section 21](#)), which lets protons pass through the barrier rather than over it.

32.2 The proton–proton chain

The *proton–proton chain* is the sequence of reactions that builds one helium nucleus from four protons, releasing energy at each step. It is the main energy source of the Sun and similar stars, and operates in stellar cores at temperatures of a few to a few tens of millions of kelvin. In its commonest branch:



which overall converts four protons into one ${}^4\text{He}$ nucleus, two positrons and two neutrinos, releasing about 26.7 MeV. The first step is the bottleneck, and it is what makes the Sun burn slowly enough to have a long life: it requires one of the two protons to turn into a neutron by the weak interaction ([Section 2](#)) at the very moment of contact, which is an exceedingly improbable coincidence. The neutrinos escape immediately and are the only direct evidence we have of the core, while the photons take a very long time to random-walk their way out. In stars appreciably more massive than the Sun, and hence hotter, the *CNO cycle* takes over instead, using carbon, nitrogen and oxygen as catalysts to achieve the same net conversion.

33 Classification of elementary particles; quarks and leptons and the particles transmitting the interactions, the structure of hadrons

The building blocks of matter are the *leptons* and the *quarks*, both spin- $\frac{1}{2}$ fermions. Leptons can exist on their own, whereas quarks are always found bound inside composite particles (hadrons). The interactions between them, their carriers and their ranges are the subject of [Section 2](#).

The leptons come in six types (three charged, three neutral), plus their antiparticles:

- the electron and the electron neutrino,
- the muon and the muon neutrino,
- the tau and the tau neutrino.

The three charged leptons carry electric charge -1 (their antiparticles $+1$), while the three neutrinos are electrically neutral. All of them feel the weak interaction.

The quarks likewise come in six types ('flavours'), plus antiquarks:

- up (u), charge $+\frac{2}{3}$,
- down (d), charge $-\frac{1}{3}$,
- strange (s), charge $-\frac{1}{3}$,
- charm (c), charge $+\frac{2}{3}$,

- bottom (b), charge $-\frac{1}{3}$,
- top (t), charge $+\frac{2}{3}$.

Besides the electromagnetic and weak interactions, quarks also feel the strong interaction, because they carry *colour charge* (labelled red, green, blue, names with no connection to optics). The carriers of the strong force are 8 massless *gluons*, which themselves carry colour and therefore interact with one another.

A hadron must be colour-neutral overall. This is achieved either by combining three quarks of three different colours (a *baryon*, such as the proton or neutron) or a quark with an antiquark of the matching anticolour (a *meson*). A useful bookkeeping quantum number is the *baryon number* (+1 for baryons, 0 for mesons). The proton is the lightest baryon, which is why the search for possible proton decay is of such interest.

34 Band structure of electronic states in crystals; metals, semiconductors, insulators

Band theory is the quantum-mechanical account of electrical conduction. Unlike the classical picture, it starts from the wave nature of electrons and from Fermi–Dirac statistics (Eq. 67). Its central idea is the *energy band*: a continuous range of energies that electrons in a solid may occupy.

The bands arise in the same way as molecular orbitals do (Section 20), only carried to the limit. Bring two atoms together and each atomic level splits in two. Bring 10^{23} together in a crystal and each level splits into 10^{23} levels so closely spaced as to form a continuous band. Between the bands lie *forbidden gaps* in which no electron state exists at all, and these are a consequence of the periodicity of the lattice: an electron wave whose wavelength matches the lattice spacing is Bragg-reflected by the crystal (Eq. 60) and cannot propagate through it.

For the outer (valence) electrons two bands matter:

- the **valence band** — the energies of electrons still bound to their atoms,
- the **conduction band** — the energies of electrons freed from their atoms, which can then carry current.

Whether a solid conducts depends not on how many electrons it has but on whether a band is partly empty. In a completely full band every state is occupied, so an applied field has nowhere to move an electron to: for every electron it would accelerate one way there is another going the other way, and the two cancel exactly. Only a partly filled band leaves states free just above the occupied ones, and only then can a field

produce a net drift. The energy dividing occupied from empty states is the *Fermi level* (Section 28), and what matters is whether it falls inside a band or inside a gap.

The size of the gap sorts materials into three classes:

- **Metals** have no gap: either the valence band is only partly filled, so its electrons move freely, or the valence and conduction bands overlap. Either way the Fermi level lies within a band, there are many mobile electrons and current flows easily.
- **Semiconductors** have a small gap (for example Ge $\simeq 0.7$ eV, Si $\simeq 1.1$ eV, GaAs $\simeq 1.4$ eV). At absolute zero the conduction band is empty and the material is an insulator, but at higher temperatures electrons are thermally excited across the gap, leaving mobile ‘holes’ behind in the valence band, so that conductivity rises with temperature — the opposite of a metal (Section 12). Semiconductors can also be *doped*: adding donor atoms with one valence electron too many gives an n-type semiconductor, adding acceptor atoms with one too few gives a p-type one. Joining the two makes a p–n junction, which passes current in one direction only, and that junction is the basis of the diode, the transistor and the solar cell.
- **Insulators** have a large gap, conventionally above about 3–4 eV (diamond has 5.5 eV), so almost no electrons reach the conduction band and current barely flows.

There is no sharp dividing line between a small-gap semiconductor and a large-gap insulator, and the boundary quoted above is only a convention — some texts put it as high as 6 eV. The gap also fixes the optical behaviour, since a photon can be absorbed only if it has at least the gap energy: this is why silicon is opaque and diamond, whose gap exceeds the energy of visible light, is transparent.